

Synthetic auxotrophy reveals metabolic regulation of plasma cell generation, affinity maturation, and cytokine receptor signaling

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eLife Assessment

The authors addressed an **important** biological question, namely the role of glutamine metabolism in humoral responses, and they obtained **solid** conclusions. The strength of this study is that the authors used state-of-the-art transgenic mouse models together with in vitro analysis, thereby providing significant insights into the question posed. The following would strengthen the manuscript: i) adding more in-depth functionality/physiological relevance in the discussion part, and ii) regarding the experiments, the inclusion of more appropriate controls and a clearer and more accurate description of the methods.

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Abstract

The efficiencies with which activated B lymphocytes proliferate and develop into antibody (Ab)- secreting plasma cells are critical determinants of adaptive humoral immunity and central to sustaining certain autoimmune diseases. Increasing evidence indicates that specific pathways in intermediary metabolism, or their substrate supply, influence lymphocyte differentiation and function. We now show that although stringent restriction of glutamine supply decreases proliferation and differentiation of B cells into plasma cells, glutaminolysis - a major means of metabolism of this amino acid - was only conditionally crucial in B cells and the Ab responses derived from them. Strikingly, *Gls*, the gene encoding the main glutaminase of lymphocytes, promoted anti-NP Ab responses at the primary and recall phases

only when either glucose uptake into B cells or pyruvate into their mitochondria was also impaired. This synthetic auxotrophy involved support to a progressive expansion of mitochondrial respiration followed by plasma cell differentiation. Surprisingly, impairment of glutaminase and the mitochondrial pyruvate channel not only decreased the coupling of IL-21 stimulation to STAT3 induction, but also interferon stimulation of STAT1 activation. Together, our findings establish not only a powerful collaboration of metabolic pathways in promoting increased respiration and the development of Ab-secreting cells, but also a capacity of metabolism to modulate cytokine receptor signaling.

Introduction

Circulating antibodies (Ab) are crucial mediators of protection against potentially pathogenic microbes, but Ab also can cause tissue damage or cellular dysfunction (1–5). In both adaptive and pathological settings, Ab functions depend on their heavy chain class, affinity, and concentration [reviewed in (6–8)]. Increased affinity after initial and repeated immunizations can be selected in several ways, one of which involves processes that occur in the micro-anatomy of germinal centers (also known as secondary follicles) [reviewed in (9–11)]. Progeny cells that survive a gauntlet of programmed death in selecting for continued clonal proliferation ultimately can generate either memory B cells (MBC) or an end-stage descendant of the B lineage, the plasma cell (PC) (11–15). The magnitude of PC output (the numbers generated after B cell activation) and their rates of immunoglobulin secretion (15–18) determine the net output and steady-state concentrations of Ab. Accordingly, understanding mechanisms that regulate the qualities or quantities of Ab, and that reduce or enhance the efficiency of generating PC, are important goals.

To elicit protective Ab, immune exposures such as vaccination activate B cell proliferation, and some progeny of such activated B cells develop into Ab-secreting PC (19–21). Specificity is fostered by a reliance on signal initiation through the BCR, the clonal Ag receptor on mature B lymphocytes (22–24). The capacity to be activated, survival efficiency, rates of proliferation, and propensity to yield GC B cells or PC depend on an array of other signal-initiating receptors, among them CD19, co-stimulators such as CD40 [reviewed in (25, 26)], the death receptor Fas (26), pattern recognition receptors in the TLR family [(27); reviewed in (28, 29)] and cytokine receptors (30–34). One consequence of activation is that B lymphoblasts dramatically increase expression of an array of nutrient transporters that include those for amino acids and for glucose [(35, 36); reviewed in (37)]. Beyond this general picture, however, the effects or outputs of receptors likely are scalar and conditional, i.e., dependent on other signals. For instance, antigen with a high affinity and high valency of the BCR epitope signals differently from a monovalent, lower-affinity antigen [(38–43), reviewed in (44)], and may influence generation of PC. Moreover, increased expression of TLR7 or type 1 interferon (IFN) receptor stimulation each reduces the capacity of B cells to maintain self-tolerance (45–48), presumably involving changes in the quantitative balance among signal transducers and transcription factors. At a later stage, the IL-21 receptor is crucial for PC differentiation (33). Moreover, IL-21 has recently been reported to promote affinity maturation of GC-derived PC (49). Little is known about the relationships between signal strength and metabolism. In vitro experiments showed that hypoxia or genetically-mediated stabilization of hypoxia-inducible transcription factors (HIFs) blunted activation-induced increases in amino acid uptake and transporter gene expression as well as GC and Ab responses (36, 50). However, it remains unclear if metabolism can modulate signaling.

Nonetheless, a substantial and increasing body of evidence indicates that nutrient supply, uptake, and intermediary metabolism influence the properties of adaptive immunity, in part by their effects on differentiation and function of cells that mediate host responses [reviewed in (31, 51–53)]. The bulk of evidence is from analyses of the differentiation and functions of T cells,

macrophages, and dendritic cells and documents condition- and cell type-specificity of findings. Nonetheless, several analyses point to important effects on B cells as well. In addition to a function during B cell development, expression of the glucose transporter GLUT1 on mature B cells supported the initial GC size, the Ag-specific Ab responses derived from B cells, and affinity maturation (35, 54, 55).

Among nutrients, only the distribution and sufficiency of oxygen are readily measured, but such analyses consistently reveal evidence of functional hypoxia in follicles, most accentuated in GC (36, 50, 56, 57). Consistent with and potentially as a result of the hypoxia, GC B cells can exhibit higher rates of glucose uptake with several-fold increased glucose oxidation as well as utilization in the pentose phosphate pathway (36, 54-56). Moreover, the ability to feed intermediates into one-carbon metabolism via increased expression of the enzyme phosphoglycerate dehydrogenase was reported to be crucial for germinal center size and normal degrees of high-affinity antibody production (58). Cell type- and stage-specific inactivation of *Slc2a1*, the gene that encodes GLUT1, indicates that the increased glucose uptake is crucial for both GC homeostasis and the capacity of B cells to yield antibody-secreting PC (54, 55). Metabolomic analyses suggested that increases in anaplerotic conversion and usage of amino acids may be evoked in GLUT1-deficient B cells as a potential compensatory mechanism (55). In parallel, however, a remarkable degree of metabolic flexibility was suggested by the in vitro finding that although glucose is absolutely essential for PC development and Ab production in the absence of other hexoses, mannose and galactose together could substitute for glucose (54). Thus, the reduced influx and availability of glucose reduced Ab responses and affinity maturation, but the decrease in these outputs fell short of what likely could be therapeutically meaningful. Moreover, the importance of feeding the increase in B cell-intrinsic generation of serine with diversion of 3-phosphoglycerate away from generating pyruvate raised the question whether the capacity of mitochondria to use this latter metabolite was important in promoting greater quality and quantity of Ab after immunization.

To explore these issues, we used both immunizations and in vitro analyses that leveraged genetic and pharmacological approaches to test the importance of glutaminolysis - on its own or in combination with a second pathway - in Ab responses. Glutamine concentrations in vitro were found to be crucial for class switching as well as PC and Ab production, and B cell type-restricted inactivation of the *Gls* gene, which encodes the glutaminase expressed in post-natal cells outside the liver, impaired the response to ovalbumin in one immunization model. Surprisingly, however, the anti-NP response in a hapten-carrier model was unaffected. However, a synergistic impact resulted when GLS depletion was combined with targeted loss-of-function for either GLUT1 (*Slc2a1*) or an essential subunit of a mitochondrial pyruvate import carrier [*Mpc2*, (59, 60)]. Thus, glutaminolysis promoted the anti-NP response only if glucose intake or pyruvate flux was impaired, revealing one limit imposed on the metabolic flexibility of B cells in their mediation of Ab responses. The combined restriction of two pathways that feed mitochondrial metabolism represented what can be termed "synthetic auxotrophy" rather than synthetic lethality (61, 62), in that effects on differentiation were observed independent from death or proliferation. Gene expression signatures in GC B cells highlighted a requirement for GLS in metabolic reprogramming, i.e., both amplification of glucose-stimulated proton secretion and a major increase in mitochondrial respiration between the first and second days after B cell activation. Surprisingly, MPC2 and GLS were found to promote not only IL-21-stimulated STAT3 activation but also an interferon response signature in addition to promoting increased respiration. Together, the results reveal that an interplay between two metabolic pathways unexpectedly converged on programming not only IL-21R activation of STAT3, PC differentiation and high-affinity Ab responses, but also the tyrosyl and serine phosphorylation of STAT1 induced by type 1 and type 2 IFNs. These findings have important implications for diseases, such as systemic lupus erythematosus, in which B cells and their response to IFN are crucial elements of pathogenesis.

Results

Alpha ketoglutarate can alter threshold glutamine concentration requirements for B cell responses

B lymphoblasts subjected to persistent stabilization of the hypoxia-inducible transcription factors took up amino acids at lower rates *in vitro* and exhibited decreased mTORC1 activity (36). mTORC1 promotes Ab class switching by a mechanism beyond effects on the rate of proliferation (36, 63, 64), and extracellular glutamine influences class switch recombination in B cells through a mechanism involving the action of mTORC1 on proteins regulating translation (65). Other work has provided evidence that plasmablasts developing in the setting of plasmodial infection are glutamine avid (66). The admixture of red and white pulp in spleen preclude accurate measurement of amino acid concentrations surrounding B cells in lymphoid follicles. Instead, we adapted a technique used for tumor masses (67, 68) to measure glutamine in the fluid eluted from transected lymph nodes, revealing a strikingly lower concentration than in plasma (Fig 1a). This apparent glutamine gradient was not a universal feature of the lymph node, in that neither leucine nor pyruvate differed substantially between plasma and interstitium (Fig 1a). To test the dependence of B cells on extracellular glutamine concentration *in vitro*, we measured both class switching and plasma cell differentiation. Titrations of glutamine in media showed that PC differentiation as well as the Ab class switch to IgG1 were impaired, but with different dose-response curves for the inflection point at which impairment was observed (addition of 0.2 mM for PC, at which the %IgG1⁺ was not affected) (Fig 1b; Supplemental Fig 1a). Although not pursued herein, it is intriguing that some aspects of the immune micro-environment appear to support robust switching and PC generation at a glutamine level that would be insufficient *in vitro*, suggesting that the required concentration is affected by the signals or nutrients *in vivo*.

Among uses of glutamine after its uptake into cells, glutaminolysis is the first step in generation of α -ketoglutarate (α KG) from the amino acid for usage in the Krebs (TCA) cycle (69). To screen whether this process might be of functional significance in the molecular regulation of the class-switch or PC differentiation, and to test if the observed effects were purely division-related, we used dye partitioning *in vitro* combined with addition of dimethyl-ketoglutarate (DMK), a cell-permeable molecule that can bypass glutaminolysis yet supply α KG. The glutamine requirement for switching to IgG1 was almost completely substituted by DMK. Moreover, development of CD138⁺ cells involved a process beyond just the proliferative effect, and DMK in the same cultures effected a definite but incomplete reversion of the glutamine requirement (Fig 1c; Supplemental Fig. 1b, c). These findings suggest that although other mechanisms may also apply, the need for glutamine involves a capacity to supply sufficient α KG to the B lymphoblasts.

Glutaminolysis in B cells promotes an anti-ovalbumin Ab response

Glutamine can be taken up via many distinct transporters, and reduction in one can be compensated by increases in others (68). However, the *in vitro* data supported testing to what extent immunization-induced Ab responses are affected by glutaminolysis. To bypass potential developmental effects in the B lineage as well as the complex roles of glutamine in T cells and other cell types (70–72), we introgressed a transgene that encodes B lineage-specific expression of a chemically activated recombinase onto a *Gls* f/f background (72, 73). Inactivation of the *Gls* gene in mature B cells (Fig 1d; Supplemental Fig. 1d), hereafter termed *Gls* iB- Δ/Δ could then be initiated at different times after establishment of a normal pre-immune B lineage. Mice (all huCD20-CreER^{T2} +/-, and *Gls* +/- or f/f) in which tamoxifen was administered prior to immunization were sensitized to ovalbumin followed by antigen (Ag) challenge in the airways. Lack of GLS (Supplemental Fig. 1d) reduced class-switched GC B cells

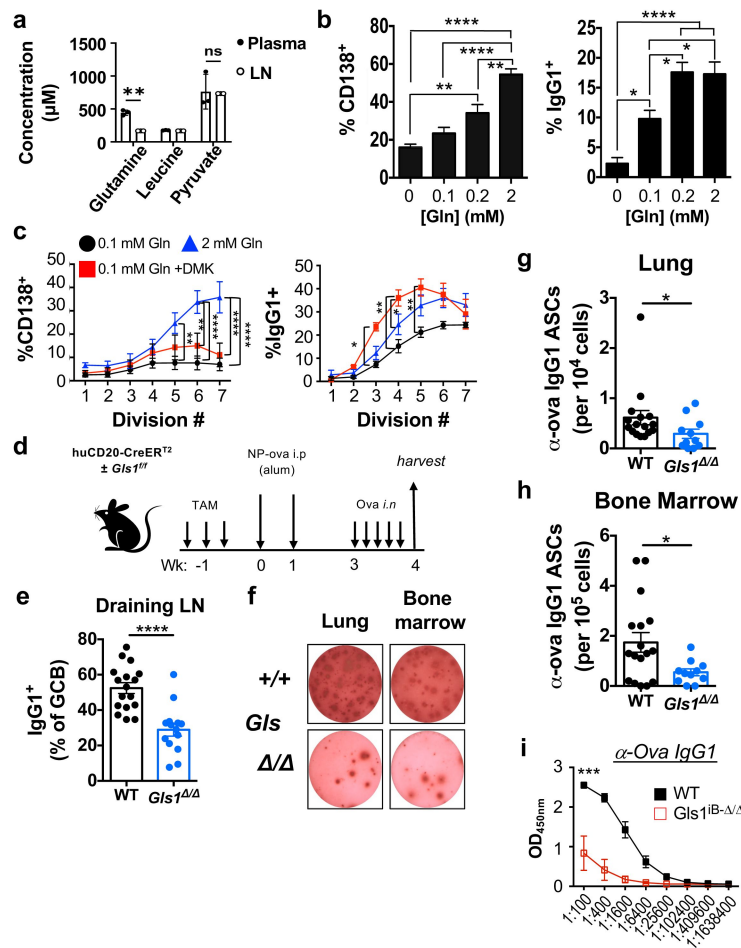


Figure 1

Glutamine and glutaminolysis promote antibody response to ovalbumin.

(a) Reduced glutamine concentration in LN relative to the bloodstream. Shown are the results from measuring concentrations glutamine, leucine, and pyruvate in the centrifugal eluates (67, 68) of lymph nodes or plasma of mice ($n=3$), as described in the *Methods*. (b) Mouse B cells were activated and cultured together BAFF, LPS, IL4, and IL5 in the presence of the indicated concentrations of Gln. Shown are the mean ($\pm$ SEM) frequencies of PC (CD138⁺ B220^{lo}; left panel) and IgG1⁺ B cells (right panel) after culture (4 d) and flow cytometry, averaging ≥ 3 independent replications. Results of measuring IgM and IgG1 by ELISA are shown in **Supplemental Fig. 1a**. (c) Reduction of PC differentiation and class switching caused by Gln restriction manifested at equal division numbers, and mitigated by a cell-permeable analogue of α -ketoglutarate. After labeling with CellTrace Violet, B cells were activated and cultured as in (b), except that dimethyl-ketoglutarate (DMK) was added to one of two cultures at 0.1 mM glutamine, as indicated. Shown are mean ($\pm$ SEM) frequencies of % CD138⁺ (left panel) and IgG1⁺ B cells (right panel) within each division-counted peak. Additional data from these experiments are presented in **Supplemental Fig 1b, c**. (d-i) An anti-ovalbumin Ab response is promoted by B cell expression of GLS. (d) Schematic of the immunization with priming and sensitization of tamoxifen-treated mice of the indicated genotypes [huCD20-CreER^{T2} and either *Gls* $+/+$ (WT) or *Gls* f/f], followed at week three by challenges with intranasal instillations of sterile ovalbumin solution. (e) Mediastinal LN were collected from harvested mice and the frequencies of IgG1⁺ events in the GL7⁺ CD95⁺ B cell gate were measured by flow cytometry. Each dot represents an individual mouse, with bars denoting the mean values. (f-h) Single-cell suspensions of lung and bone marrow were analyzed by ELISpot assays with ovalbumin-coated filters to capture secreted Ag-specific Ab detected using anti-mouse IgG1. Shown are (f) representative wells from the indicated sources (organ; genotype of B cells) and aggregated frequencies of anti-ova IgG1⁺ ASCs in lung (g) and marrow (h). (i) Anti-ova IgG1 in sera of the mice with B cell-restricted depletion of GLS (*Gls* iB- Δ/Δ) or controls, as indicated. Shown are data from ELISA with serial 4-fold dilutions using individual samples from each mouse. The likelihood of each null hypothesis (no true difference between two samples) was calculated as noted in the *Methods* (two-tailed, and non-parametric testing where conditions not met for Student's *t*-test). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

in lymph nodes draining the lung (**Fig 1e**) as well as the populations of ovalbumin-specific Ab-secreting cells (ASCs) in the lung and bone marrow (**Fig 1f-h**). In line with these data, anti-ovalbumin IgG1 and IgM concentrations in sera of *Gls* iB-Δ/Δ mice were ~0.25x those of CreER^{T2} controls (**Fig 1i**; **Supplemental Fig. 1e**). We conclude that B cell glutaminolysis can promote the development of Ag-specific ASCs and an anti-protein Ab response.

Glutaminolysis is dispensable in an anti-NP response yet synergizes with other metabolic needs

Mitochondrial oxidative metabolism appears to promote affinity maturation in the classic hapten-carrier model of an Ab response specific for nitrophenolic (NP) adducts (74-76). In contrast to the impact of *Gls* inactivation on an anti-protein response (**Figs 1**, **2a-c**), the anti-NP Ab response and affinity maturation were unaffected (**Fig 2a, d-i**). However, inducing B cell-specific inactivation of *Gls* synergized with loss of GLUT1. Thus, the combination of inducible B cell-specific gene targeting (iB) that restricted glucose import into B cells (35, 54, 55) with GLS depletion caused dramatically greater decreases in anti-Ova and anti-NP ASCs than iB inactivation of *Slc2a1* alone (**Fig. 2b-f**). Combined *Gls* and *Slc2a1* inactivation had a relatively modest impact on IgM anti-NP compared to iB-*Slc2a1* Δ/Δ samples (**Fig 2g**). In contrast, a synergism was evident in the effects on circulating Ag-specific IgG1, for which iB-inactivation of both *Gls* and *Slc2a1* dramatically compromised the humoral response, whereas GLUT1 depletion reduced anti-NP Ab to 1/2 - 1/4 levels of WT controls and a single loss-of-function mutation for *Gls* had no effect (**Fig 2h-i**). The synergism was most striking for high-affinity IgG1 (**Fig 2i**), though it also impacted all-affinity IgG1 (**Fig 2h**).

Glucose oxidation rates are several-fold higher in GC B cells than in their naive counterparts (54). However, the effects of eliminating GLUT1 on a GLS-deficient background could be due to glucose usage in serine generation (58), the pentose phosphate pathway (54), or glycosylation (60) rather than from reduced pyruvate flow into mitochondrial metabolism. Directly to interfere with pyruvate import into mitochondria for generation of acetyl-CoA, we used a previously characterized allele for inducible inactivation of *Mpc2*, a gene encoding an essential subunit of the mitochondrial pyruvate channel (**Fig 3a**) (59, 60). Levels of class-switched anti-NP antibodies in sera after the second immunization were modestly reduced after inducible *Mpc2* targeting in B cells (**Fig 3b, c**), as was affinity maturation (**Fig 3d**). Whereas GLS depletion had no effect on its own, the combined loss-of-function synergized and dramatically reduced Ag-specific IgM (**Supplemental Fig. 2a**) and responses of several switched isotypes (**Fig 3b, c**). Moreover, affinity maturation was reduced further (**Fig 3d**), and high-affinity IgG2c anti-NP in mice with doubly deficient B cells was ~ 0.02x controls while the all-affinity was ~ 1/8th of the control values (**Fig. 3c**). B cells with compound loss-of-function yielded dramatically fewer cells secreting high-affinity anti-NP Ab (**Fig. 3e-g**), whereas for GC- and memory-phenotype B cells (GCB, MBC) the magnitude of decrease was modest (~ 1/2 control numbers; **Supplemental Fig 2b-d**) in parallel with decreased BrdU uptake into GC B cells (**Supplemental Fig. 2e**). In contrast to our findings with conditional targeting of *Slc2a1* in B cells, in which a substantial degree of counterselection was evident (54), the mRNA encoding both *Gls* and *Mpc2* were far lower in flow-sorted GC B cells (**Supplemental Fig 2f, g**).

Most of the splenic ASC pool at 7 d after a recall challenge and high-affinity IgG anti-NP Ab in sera is expected to derive from re-activation of memory B cells (MBCs) that directly differentiated into PCs after the second immunization (14, 77, 78; reviewed in 11, 15). We tested how the loss-of-function interventions impact the recall process by performance of a priming immunization, tamoxifen treatments only several weeks thereafter, followed by the second immunization and harvest (**Fig 3h**). In this setting where the primary response from a pre-immune repertoire was unperturbed, a synergistic interplay of *Gls* and *Mpc2* was again observed (**Fig 3i**; **Supplemental Fig. 3a-h**). Specifically, there were substantially decreased frequencies of anti-NP ASCs (**Supplemental Fig. 3g, h**). Moreover, high-affinity anti-NP Ab of several

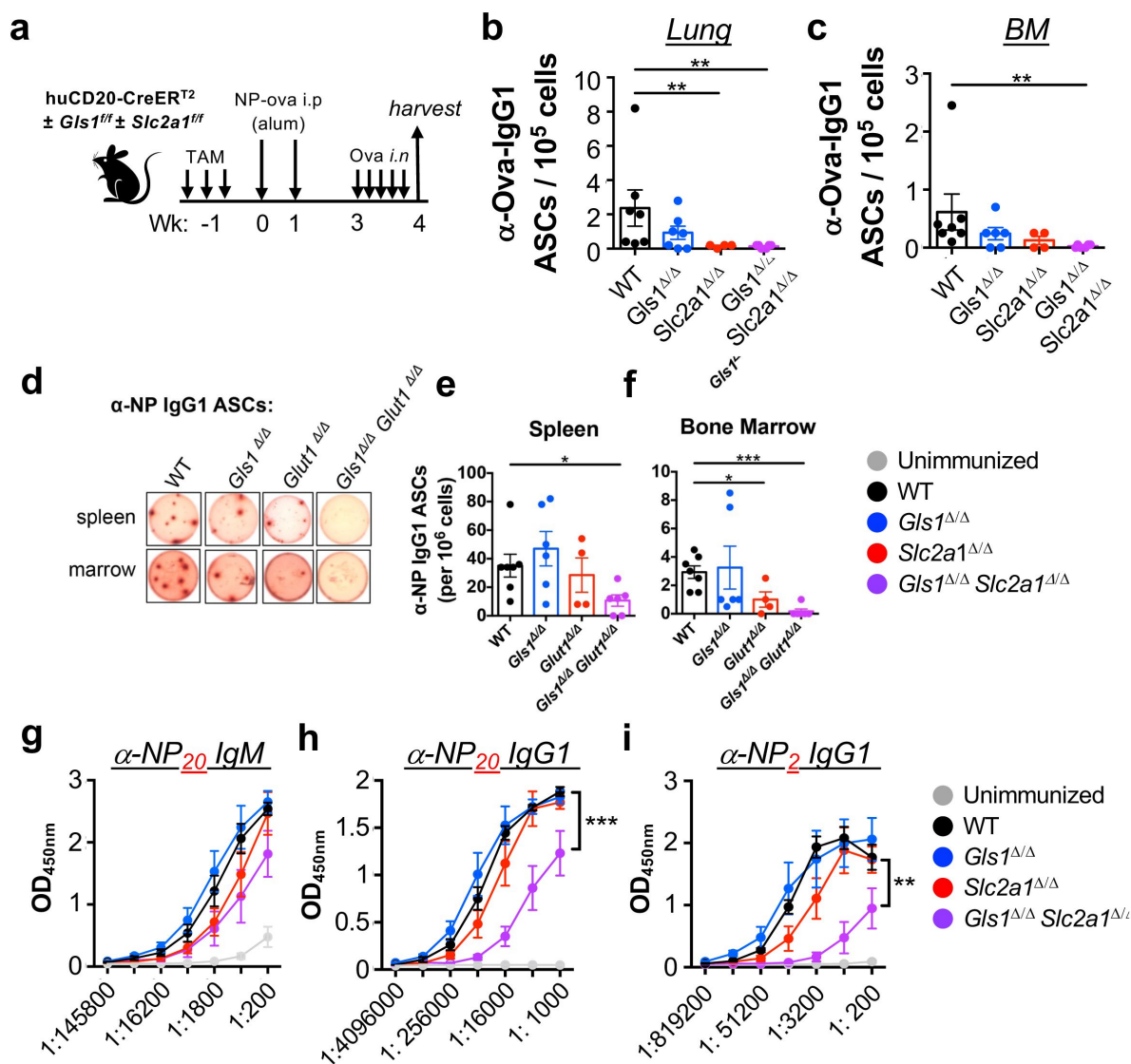


Figure 2

Glutaminolysis only conditionally supports the anti-NP Ab response.

(a) Schematic of the immunization with priming sensitization and inhaled challenge of tamoxifen-treated mice of the indicated genotypes [huCD20-CreER^{T2} and the indicated combinations of either *Gls* +/+ (WT) or *Gls* f/f and either *Slc2a1* +/+ (WT) or *Slc2a1* f/f], followed at week three by challenges with intranasal instillations of sterile ovalbumin solution. (b-f) ELISpot analyses of cells secreting IgG1 anti-ovalbumin (b, c) and anti-NP (d-f) Ab in mice of the indicated genotypes. Shown are counts from the lung suspensions (b), bone marrow (c, d, f) and spleens (d, e) from each individual mouse, with means (±SEMs) for the aggregate data denoted as bar graphs. (d) Representative ELISpot wells scoring the frequencies of IgG1 anti-NP Ab-secreting cells in spleen and marrow of mice whose B cells were converted to the indicated genotypes [wildtype (●), *Gls*^{Δ/Δ} (●), *Slc2a1*^{Δ/Δ} (●), and *Gls*^{Δ/Δ}*Slc2a1*^{Δ/Δ} (●)] after immunization and rechallenge. Data pooled from two independent experiments (seven wildtype, six *Gls*^{Δ/Δ}, four *Slc2a1*^{Δ/Δ}, and six *Gls*^{Δ/Δ}*Slc2a1*^{Δ/Δ} mice. * p < 0.05, ** p < 0.01, *** p < 0.001 (Mann-Whitney U test). (g-i). Shown are the mean (±SEM) absorbances of aggregated results of ELISA analyses of the IgM (g) and IgG1 (h, i) responses in the immunized mice of the two independent experiments, comparing all-affinity (h) to high-affinity (i) anti-NP, using serial 3-fold or 4-fold dilutions when detecting IgM and IgG1, respectively.

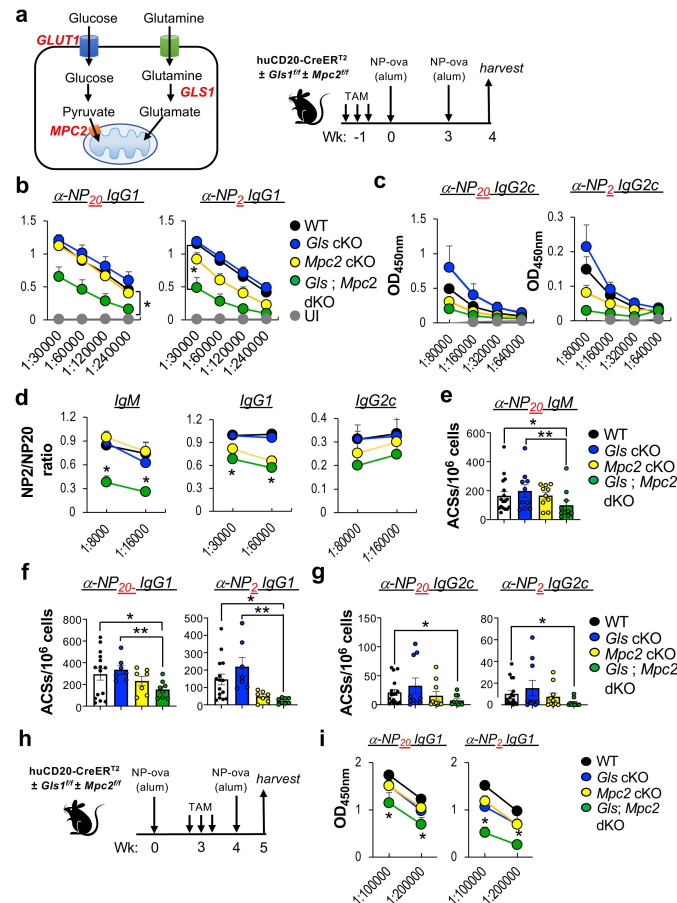


Figure 3

Synthetic auxotrophy - glutaminase support of anti-NP response is dependent on mitochondrial pyruvate channel subunit 2.

(a) Schematic of the immunization with priming and a recall boost of mice of the indicated genotypes [*huCD20-CreER^{T2}* and the indicated combinations of either *Glis* +/+ (WT) or *Glis* f/f and either *Mpc2* +/+ (WT) or *Mpc2* f/f], treated with tamoxifen prior to the initial immunization. (b-d) Results of serological analyses performed on sera from the time of harvest. Shown are the mean ($\pm$ SEM) absorbances measured in ELISA for detection of all- and high-affinity Abs (α -NP₂₀ and α -NP₂, respectively) of the (b) IgG1 and (c) IgG2c isotypes. Serologic measurements of IgM anti-NP Ab are shown in **Supplemental Data Fig. 2a**. (d) Impacts of altered metabolic pathways in B cells on relative extents of affinity maturation in the recall responses. Ratios of absorbances are shown for high- (NP₂) / all-affinity (NP₂₀) ELISA at two serum dilutions for the IgM, IgG1, and IgG2c isotypes, as indicated. (e-g) Shown are mean ($\pm$ SEM) frequencies of splenocytes producing anti-NP Ab of the indicated isotypes [(e) IgM, (f) IgG1, (g) IgG2c] and affinities at harvest after the booster immunization, as in (a), with genotypes as shown and each dot representing an individual mouse. (b-g) Results are aggregated from four temporally separate immunization experiments with mice of each genotype, totaling eight individual controls [(WT, i.e., only *CreER^{T2}*) (●)] and eight mice with each type of induced B cell type-specific deletion [*Glis^{Δ/Δ}* (●), *Mpc2^{Δ/Δ}* (●), and *Glis^{Δ/Δ}, Mpc2^{Δ/Δ}* (●) genotypes]. **Supplemental Data Fig. 2** shows additional results, with frequencies of GC B cells, MBCs, BrdU uptake rates in GC B cells, as well as *Glis* and *Mpc2* mRNA. (h, i) Functional requirements for GLS and MPC2 in B cells during the recall phase. (h) Schematic of the priming immunization without deletion of the conditional alleles, followed by tamoxifen injections ~ 3 wk later, and only then a recall boost of mice of the indicated genotypes [*huCD20-CreER^{T2}* and the indicated combinations of either *Glis* +/+ (WT) or *Glis* f/f and either *Mpc2* +/+ (WT) or *Mpc2* f/f]. (i) Shown are serologies of the all- and high-affinity IgG1 anti-NP Ab elicited by a boost when Cre activation was initiated only after the primary response. Results are aggregated from three temporally separate immunization experiments with mice of each genotype, totaling nine individual controls [(WT, i.e., only *CreER^{T2}*) (●)] and nine mice with each type of induced B cell-specific deletion [*Glis^{Δ/Δ}* (●), *Mpc2^{Δ/Δ}* (●), and *Glis^{Δ/Δ}, Mpc2^{Δ/Δ}* (●) genotypes]. **Supplemental Data Fig. 3** shows additional results, with IgM, IgG2c and IgA isotypes; frequencies of ASCs and NP-binding GC B cells.

switched isotypes were drastically reduced: IgG1 to $< 1/4$ of control values (**Fig 3i**), with IgG2c and IgA almost undetectable (**Supplemental Fig 3a-c**). Since high-affinity, class-switched (IgG or IgA) Ab are undetectable 1 wk after a primary immunization, we infer from these results that the rapid generation of ASCs from memory B cells required glutaminolysis and pyruvate import. Together, the findings reveal a cell-intrinsic process that promotes both affinity selection and differentiation toward a PC fate through the collaboration of pyruvate influx rate and glutaminolysis.

Synthetic auxotrophy of progressively increased respiration and PC differentiation

Glutaminolysis and mitochondrial pyruvate import each can be inhibited by a well-characterized pharmaceutical compound - CB839 (72) and UK5099 (59, 60), respectively. We tested the effect of these agents - in combination, or each one singly - on physiology and differentiation of B cells activated via CD40. CB839 reduced the production of CD138⁺ progeny (**Fig 4a, b**), ASC (**Fig 4c, d**) and Ab (**Fig 4e**). Consistent with its ability to inhibit glutaminase, this agent decreased intracellular glutamate and the calculated glutaminolysis rate while increasing the glutamine pool (**Supplemental Fig. 4a-c, e**). On its own, UK5099 had no effect on development of CD138⁺ cells, yet addition of UK5099 to CB839 substantially intensified the inhibition of ASC generation and Ab production. Of note, UK5099 reduced neither glutamine nor glutamate, yet substantially reduced intracellular alanine, which is generated in mitochondria by glutamate-pyruvate transaminase 2 (GPT2) conversion of glutamate and pyruvate to alanine and α KG (**Supplemental Fig. 4b, d**). Combining UK5099 with CB839 trended toward accentuation of the glutamate reduction caused by the glutaminase inhibitor on its own (**Supplemental Fig. 4b, c**). Further, the frequencies of ASCs and spot sizes were substantially reduced when metabolism was inhibited only during the differentiation phase, followed by washout of the inhibitors (**Supplemental Fig 5a, c**). These data indicate that glutaminolysis and mitochondrial pyruvate import promote the differentiation process, but results from applying inhibitors only during the ELISpot culture, i.e., after 4 d of differentiation *in vitro*, suggest that glutaminolysis also non-redundantly supports the synthesis and/or secretion of Ab (**Supplemental Fig 5b, d**).

Progression from B cell activation to the differentiation of plasmablasts and plasma cells requires several rounds of cell cycling and a division-counting mechanism (79). When the frequency of CD138⁺ progeny was measured as a function of division number, the results showed that CB839 reduced the capacity to become CD138-positive at equal divisions (**Supplemental Fig 6a-d**). Moreover, whereas UK5099 on its own did not reduce differentiation efficiency at equal divisions, its addition intensified the inhibitory effect of CB839 (**Supplemental Fig 6a-d**). Recent work with profound B cell-depletion therapy in chronic autoimmune disease indicates that disease activity depends on continual production of new auto-Ab-producing cells [80, reviewed in (81)]. Hydroxychloroquine (HCQ) is the standard of care in systemic lupus erythematosus; while effective, it incompletely reduces pathogenic auto-Ab concentrations and disease activity, and often is not sufficient (82). We explored the interplay between HCQ and CB839 (glutaminase inhibition), alone or along with UK5099. Strikingly, whereas HCQ alone slightly increased the %CD138⁺ at each division and overall, addition of HCQ to metabolic inhibition caused more substantial reductions in PC differentiation (**Supplemental Fig 6a-d**).

Levels of *Prdm1* mRNA, which encodes a transcription factor that determines PC fate and identity [reviewed in (9, 83)], were lower in inhibitor-treated cells at a time in the cultures when the CD138⁺ population just started to develop (**Fig 4f**). This finding is consistent with an effect on differentiation beyond simple division-counting and with the pattern of results with metabolic inhibition. Moreover, analyses of cell-type modules in RNA-seq data from B lymphoblast cultures at this inflection point yielded evidence that the main systemic impact of combined inhibition was on plasmablast - plasma cell gene expression programs (**Fig 4g, h**). We further found that DMK, the cell-permeable α KG precursor, partially reversed the inhibitory effects of CB839 and UK5099

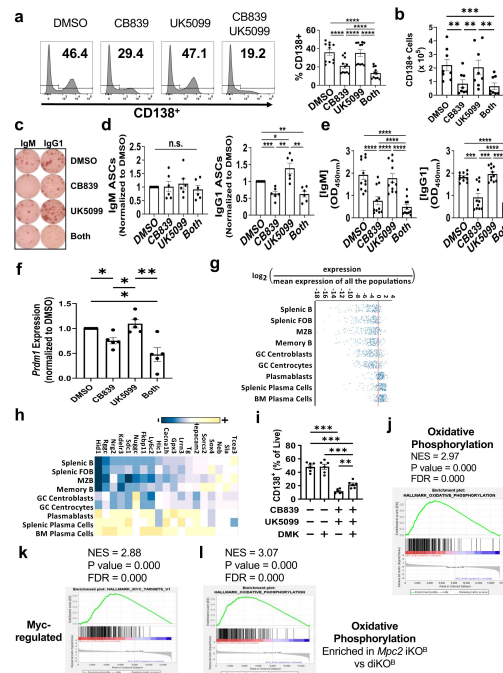


Figure 4

GLS and MPC2 collaborate in supporting progression to plasma cell development.

(a) B cells were activated and cultured under conditions promoting plasma cell differentiation in the presence of the indicated combinations of vehicle (DMSO) or inhibitors of GLS (CD839) and the MPC (UK5099). Shown are representative histograms from flow cytometric analyses of CD138 within the live cell gate, with inset numbers denoting the %CD138⁺. The bar graph shows the mean (±SEM) results for generation of CD138⁺ cells under each treatment condition, pooling data from three temporally independent experiments, each with 3-5 independent B cell pools purified from separate mice (each dot represents a distinct sample). (b) Shown are the mean (±SEM) calculated numbers of PC generated in temporally independent replica experiments with a total of eight independent B cell pools cultured in vitro in (a). (c) Representative ELISpot results measuring the frequencies of IgM- and IgG1-secreting PC, as indicated, produced in the differentiation cultures under each treatment condition. (d) Bar graphs with mean (±SEM) ELISpot results pooled from the replicate experiments illustrated in (c), with each dot representing an individual sample. Shown are data normalized to the vehicle (DMSO) control for each set of cultures using an individual B cell pool. (e) Bar graphs show the mean (±SEM) absorbance values from ELISA measurements of IgM and IgG1 secreted into the media during the cultures as in (b). Additional data quantifying ASCs are presented in [Supplemental Fig. 4](#) [\[4\]](#). (f) *Prdm1* gene expression promoted by GLS and MPC2. B cells were activated and cultured as above, but harvested after 3.5 d culture in BAFF, IL-4, IL-5, and the indicated inhibitor(s) or vehicle followed by qRT2-PCR to quantitate *Prdm1* RNA encoding Blimp1. Shown are the results from four biologically independent mouse pools, B cell purifications and cultures, with the *Prdm1*-encoded RNA then normalized in each experiment to the level in the vehicle (DMSO) control (in each sample, relative to the averaged C_T values of cyclophilin A and GAPDH). (g, h) Global gene expression identifies plasma cell identity as a main target of synthetic auxotrophy. Using three biologically independent replicate pools for each condition, RNA-seq was performed with the B cells cultured as in (f). Enriched genes identified by DESeq2 comparison were analyzed using the MyGeneset tool from ImmGen. (g) Genes enriched in vehicle treated cultures compared to cultures treated with both CB839 and UK5099 are shown as a W-plot with defined stages for mature B cells and PC indicated. (h) Genes enriched in CB839-treated cultures compared to cultures treated with both CB839 and UK5099 are shown as a heatmap of z-scored relative expression, with specific gene identities and defined stages for mature B cells and PC indicated. (i) Metabolic mitigation of the block imposed by synthetic auxotrophy. Graphs of aggregate results from six biologically independent B cell preparations (two biological replicates in each of three independent experiments), presented as in (a), are shown for differentiation assays performed with B cells purified, activated, and cultured as in (a), except that the cell permeable αKG analogue DMK was added as indicated. (j-l) Gene set enrichment analyses were performed on RNA-seq data generated using RNA from flow-purified GC B cells and hallmark gene sets of the Mouse Molecular Signatures Database. Shown is a selected subset of analyses with high normalized enrichment scores (NES) for the indicated gene sets (j) oxidative phosphorylation (WT vs *Gls* Δ/Δ, *Mpc2* Δ/Δ GC B cells); (k) regulated by c-Myc (*Mpc2* Δ/Δ vs *Gls* Δ/Δ, *Mpc2* Δ/Δ GC B cells); (l) oxidative phosphorylation (*Mpc2* Δ/Δ vs *Gls* Δ/Δ, *Mpc2* Δ/Δ GC B cells). Additional GSEA and other data are in [Supplemental Figure 7](#) [\[4\]](#).

on B cell differentiation to a CD138⁺ plasmablast/PC status (**Fig. 4i**). While not the entire mechanism, these findings indicate that αKG generation via glutaminolysis promotes PC differentiation.

The serological data highlight an impact on affinity maturation of class-switched Ab after a secondary immunization. T cell help drives greater proliferation of both extrafollicular- and GC- B cell-derived plasmablasts that bore higher-affinity BCR (49, 84), but the findings (**Fig 3b-d**; **Supplemental Fig 2c, e**) demonstrate effects on GC B cells. Since in vitro systems cannot faithfully recapitulate GC B cell physiology, we performed RNA-seq analyses of flow-purified GC B cells from the mice with induced B cell type-specific gene inactivation and matched controls (CreER^{T2} transgenics injected with tamoxifen). The analyses consistently showed major reductions in the mRNA encoded by the targeted genes (*Mpc2*; *Gls*), indicating both that deletion of the floxed alleles was efficient and that the GC were not dominated by selective outgrowth of the B cells that retained functional genes (**Supplemental Fig 2f, g**). Gene set enrichment analyses (GSEA) of the impact of GLS on the relative RNA levels in GC B cells highlighted modules for the interconnected functions of Myc-regulated genes, those connected to mTOR signaling and oxidative metabolism (oxidative phosphorylation; fatty acid oxidation) (**Fig 4j-l**; **Supplemental Fig 7**). Moreover, the abnormal metabolism led to reduced expression of RNAs linked to proliferation (E2F-responsive genes and those associated with the cell cycle G2-M transition) (**Supplemental Fig 7a**). These effects could be discerned when comparing *Gls* Δ/Δ B cells to CreER^{T2}, +/+ controls but effects were heightened when *Mpc2* inactivation was compared to *Mpc2* Δ/Δ, *Gls* Δ/Δ samples or in comparing doubly-deficient B cells to WT controls (**Supplemental Fig 7**).

The gene modules pointed to proliferation, c-Myc (a central mediator of metabolic reprogramming), and oxidative metabolism. Oxidative metabolism is vital for PC differentiation (85). Accordingly, we measured respiration by B lymphoblasts of each genotype on two successive days after activation and culture in conditions promoting development into ASC. Of note, rates of mitochondrion-dependent oxygen consumption by WT cells quadrupled between days 1 and 2 after mitogenic stimulation, (**Fig 5a-d**). Consistent with the GSEA data, the increased respiration was blunted in GLS-deficient B cells (**Fig 5a, b**). In addition, the inactivation of *Mpc2*, while not affecting respiration on its own, further reduced both the absolute rate and the capacity to increase this core mitochondrial function during the second day when GLS was deficient (**Fig 5a, b**). To explore if these effects were attributable either to abnormal naive B cells prior to stimulation, or simply to a failure to provide sufficient substrates to the mitochondria, we added CB839 or UK5099 to cultures of normal B cells after activation but performed the metabolic flux assays in the absence of inhibitors. Strikingly, the impacts on respiration caused by pharmacological inhibition initiated at the time of B cell activation matched those obtained using genetic loss-of-function (**Fig 5c, d**). Further consistent with altered mitochondrial metabolism, the inhibition of glutaminolysis also impacted the acetyl-carnitine /ratio that is central to the provision of substrates for fatty acid oxidation, with reduced L-carnitine and increased acetyl-carnitine as well as computationally inferred rates of fatty acid oxidation (**Supplemental Fig. 4g-j**).

Rates of extracellular acidification rates (ECAR) were measured to test if B lymphoblasts compensated for the reduced respiration by increasing "aerobic glycolysis" - the hallmark of which would be to generate and excrete lactate from pyruvate. Targeting glutaminolysis led to striking decreases in ECAR that were even greater when in concert with interference with the MPC (**Fig 5e, f**). These decreases indicate that the interference with glutaminolysis broadly reprograms metabolism in the activated B cells. Although activated B cells perform fatty acid oxidation [(104); reviewed in (37)] the inhibitors decreased steady-state ATP concentrations in the CD40-activated lymphoblasts (**Fig 5g**). Of note, the main cause of these decreases appeared to be interference with the capacity of B cells to increase their respiratory capacity during growth (**Fig 5h**) rather than the modest impact of acute inhibition of glutaminolysis and pyruvate import in blasts cultured in the absence of inhibitors during 2 d prior to adding inhibitors (**Fig 5i**). The

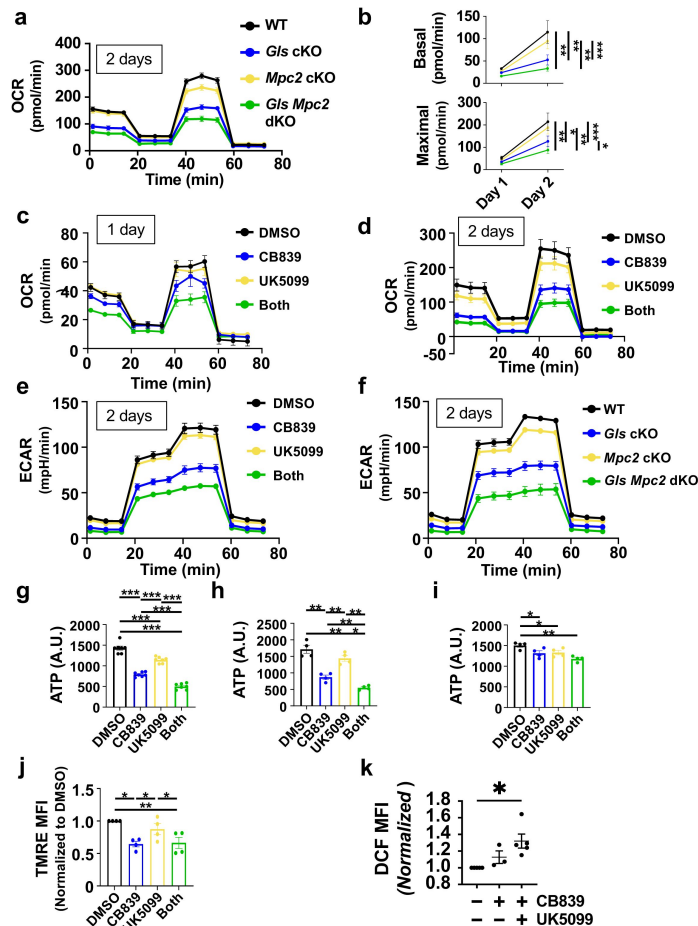


Figure 5

Synthetic auxotrophy of B cell metabolism that supports a progressive post-activation increase in mitochondrial respiration.

(a) Pools of purified B cells from WT mice or those with the indicated gene-targeted loss(es) of function were activated and cultured (1 and 2 d) with α CD40, BAFF, IL-4, and IL5. Metabolic functions were then assayed by a metabolic flux analyzer. (b-f) As for (a), except WT cells were treated with inhibitors (CB839; UK5099) alone or in combination, as indicated by the color coding, and then subjected to mitochondrial stress-tested measurements of respiration (b), biochemical assays of [ATP] (g-i), flow cytometry (j, l) or qPCR (k). (a) Oxygen consumption rates (OCR) during mitochondrial stress testing of B cells, comparing loss-of-function B cells of the indicated genotypes, color-coded as in Fig. 3. (b-d) Changes in basal respiration and maximal respiration of B cells from day 1 to day 2 after activation (b), calculated from assays in (c) and (d) OCR values at day 2 were used for statistical analysis. (c, d) As in (a) except that purified WT B cells were used, with additions of vehicle (DMSO) or the indicated inhibitor(s) (CB839, 1 μ M; UK5099, 10 μ M), with each B cell pool assayed on both days 1 (c) and 2 (d) after purification and activation. (e) Extra-cellular acidification rates (ECAR) during glycolytic stress tests of WT B cells activated and cultured in the presence of the indicated agents after 2 d cultures as in (d). (f) ECAR during glycolytic stress test of B cells inducibly rendered *Gls* Δ/Δ and/or *Mpc2* Δ/Δ , then activated and cultured as in (a). (g, h, i) Intracellular [ATP] in lysates of B cells activated and cultured (2 d) as in (c). (g) Metabolic inhibitor(s) were present throughout the period of culture (2 d) and assay. (h) Cells were activated and initially cultured in the presence of the indicated metabolic inhibitor(s), then washed, and assayed (90 min) in medium without inhibitors. (i) After activation and 2 d culture with no inhibitor present, the indicated agent(s) were added to block glutaminolysis and/or mitochondrial pyruvate import during the 90-minute assay. (j) Mitochondrial membrane potential determined by tetramethylrhodamine (TMRE) staining analyzed by flow cytometry. Shown are mean fluorescence intensity (MFI) values from each independent experiment after activation and culture (2 d) as in (c), then normalized to DMSO-treated condition in each experiment. (k) Flow cytometry results comparing inhibitor-treated cells vs controls after staining for ROS with DCFDA in three independent replication experiments, with each dot representing one experiment, normalizing as in (i).

reduced respiration, lower signals from TMRE staining (**Fig 5j**), together with normal signal after MitoTracker Green staining and qPCR for mitochondrial DNA content (**Supplemental Fig 6e, f**), indicate that decreased respiration was linked to a reduction in mitochondrial membrane potential rather than a failure to increase mitochondrial mass or DNA. This decrease in turn was associated with an increase in reactive oxygen species (ROS) (**Fig 5k**).

Metabolic regulation of interferon cytokine receptor responsiveness

To gain further insight into molecular consequences of synthetic auxotrophy, we analyzed the RNA-Seq data to determine how *Mpc2* gene inactivation affected GLS-depleted GC B cells. Surprisingly, the RNA-seq data indicated that a Jak-STAT3 signature was reduced not only in GC B cells deficient GLS and MPC2 but also in the comparison of *Mpc2* Δ/Δ to combined loss-of-function (**Fig 6a**; **Supplemental Fig 7a**). IL-21R, acting via STAT3, is crucial for PC differentiation (33, 34, 86, 87). Cytokine receptors activate STAT3 via Jak1-mediated tyrosine phosphorylation, but ROS can alter STAT3-dependent gene transcription (88–90). We found that the tyrosyl phosphorylation of Jak1 and STAT3 stimulated by IL-21 treatment of activated B cells was attenuated in cells cultured in CB839 and UK5099 (**Fig 6b**). Unexpectedly, computational algorithms identified the interferon (IFN)-stimulated gene signature as the most prominently impaired pathway in the *Mpc2*, *Gls1* double-deficient GC B cells (**Fig 6c**). Thus, the effects of MPC2 in the setting of GLS depletion were to promote interferon response signatures and Jak-STAT signaling as well as suites of RNA linked to respiration (**Fig 6d**; **Supplemental Fig 7a, d**). Among processes that could lead to such a strong effect on mRNA levels, a straightforward possibility was that the metabolism in B cells affects their IFN-elicited signal transduction. Strikingly, IFN-induced phosphorylation of the conserved tyrosine was attenuated when activated cells were treated with CB839 and UK5099 prior to stimulation with type 1 or 2 IFN (IFN- β and - γ , respectively) (**Fig 6e, f**). This effect was observed in B cells three days after initial activation via CD40 crosslinking, and inhibition was observed when the pharmacological agents were present only for the final 16 h, or when anti-CD40-activated blasts were analyzed the day after activation (**Fig. 6e, f**). Decreased IFN-stimulated phospho-STAT1 was also observed in comparisons between WT and *Gls* Δ/Δ *Mpc2* Δ/Δ B cells from tamoxifen-treated mice (**Supplemental Fig. 8a–e**). One consequence of STAT1 nuclear translocation and chromatin association is that the C-terminal transcription activation domain (TAD) can become phosphorylated at S727, an event that depends on the Jak/Tyk2 tyrosyl kinase action (91). Of note, IFN-induced STAT1(Y727) phosphorylation also was attenuated when GLS and MPC2 were impeded in activated B cells (**Fig. 6e, f**). We conclude that altered intermediary metabolism in B cells can, within less than a day, change IFN receptor induction of phospho-STAT1.

Finally, we tested models of mechanisms that could mediate the observed ability of metabolism (i.e., glutaminolysis and mitochondrial pyruvate import) to increase STAT1 phosphorylation in activated B cells. Flow cytometry results indicated that a simple model in which surface expression of IFN receptors was reduced is unlikely (**Supplemental Fig 8f**), suggesting instead an impairment of receptor signaling to STAT1. One observed effect of the combined metabolic reprogramming was to increase steady-state ROS in B cells (**Fig 5k**). Of note, ROS can reduce function of complex I of the electron transport chain (ETC) by modifying reactive sulphydryl groups in the complex, and elimination of complex I from the mitochondrial electron transport chain (ETC) decreased IFN- γ activation of STAT1 in a genome-wide loss-of-function screen with an immortalized macrophage-like cell line (92). The absence of complex I was mimicked by the ETC inhibitor rotenone (92). Longer rotenone treatments were highly toxic to B cells, but we found that p-STAT1(Y701) induction in activated B cells was dramatically decreased by a remarkably short (45 min) exposure to this agent prior to IFN stimulation (**Fig. 6g, h**). Metformin also inhibits complex I, albeit at higher concentrations than apply clinically (93, 94). When tested with activated B cells, this pharmaceutical agent also decreased p-STAT1 elicited by IFN stimulation (**Fig. 6g, h**).

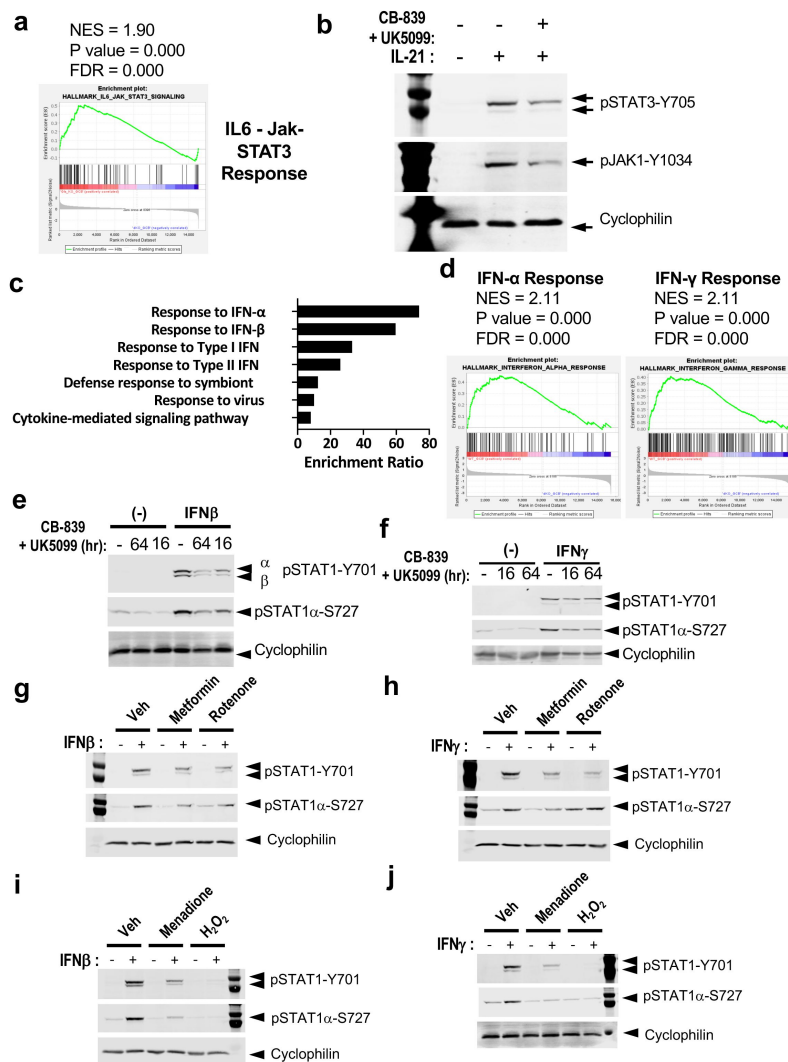


Figure 6

Metabolism in B cells promotes interferon receptor signaling to STAT1.

(a) Enrichment of the IL6-stimulated Jak-STAT3-induced gene set. (b) Immunoblot analyses of IL-21-induced tyrosine phosphorylation of STAT3 in B cell blasts, showing representative results from one individual experiment representative of three independent replications. Purified B cells were activated with anti-CD40 and BAFF, cultured for 16 hr in the presence of vehicle (DMSO) or inhibitors (CB839 and UK5099), as indicated, then stimulated (15 min) with IL-21. (c) Results from an over-representation analysis of differentially expressed protein-coding RNA in WT versus induced double-deficient (GLS; MPC2; "diKO^B") B cells, as counted in the RNA-seq analyses with GC B cells of SRBC-immunized mice (as in **Fig 4, j-l**). Additional GSEA and an overview of gene set comparisons for pairs of genotypes are presented in **Supplemental Fig. 6**. (d) Selected analyses of gene sets enriched in WT GCBs compared to diKO^B GCBs using the Hallmark Pathway database. Shown are enrichment of IFN- α - and IFN- γ -associated pathways in WT GCBs compared to the *Glis* Δ/Δ , *Mpc2* Δ/Δ samples. (e-j) Immunoblot analyses of IFN-induced STAT1 phosphorylation in activated B cells, showing representative results from individual experiments, each representative of three independent replications. (e, f) Purified B cells were activated with anti-CD40 and BAFF and cultured for 64 or 16 hr in the presence of vehicle (DMSO) or inhibitors (CB839 and UK5099), as indicated, then stimulated (15 min) with IFN- β (e) or IFN- γ (f) followed by immunoblotting using Ab specific for p-STAT1(Y701) or p-STAT1 α (S727) along with anti-cyclophilin B Ab as a loading control. (g, h) Inhibition of mitochondrial ETC attenuates STAT1 activation. B cells were activated and cultured as in e and f but for 16 hr, in the presence or absence of metformin (2 mM; 16 hr) or rotenone (0.5 μ M; final 2 h of the cultures) as indicated, then stimulated (15 min) with IFN- β (g) or IFN- γ (h) and analyzed as for panels d, e. (i, j) ROS inhibit STAT1 activation. B cells were activated and cultured for 16 hr, in the presence or absence of menadione (2 μ M; all 16 h) or H₂O₂ (100 μ M; the final 2 h), and then stimulated with IFN- β (i) or IFN- γ (j) for 15 min.

Dysfunction or chemical inhibition of ETC complex I causes increased mitochondrial production and export of ROS (95 [↗](#)), so it was possible that these results were in part attributable to elevated ROS. Using an approach that had provided evidence of ROS inhibiting STING-dependent IFN- β production (96 [↗](#)), we increased ROS in activated B cells by applying either menadione or H₂O₂ prior to interferon treatment. Menadione substantially reduced induction of P-STAT1 by receptors for both IFN- β and - γ , which was almost completely abrogated by a short-term pre-treatment with H₂O₂ (Fig 6i, j [↗](#)). We conclude that increased steady-state ROS and mtROS in B cells, which arise when GLS and MPC function are impaired, contribute to an unexpected defect in IFN-R signaling to STAT1. All together, the results reveal that an interplay between two metabolic pathways progressively reduced the expansion of mitochondrial respiration in B cells. A key functional effect of the combined impairment of two metabolic pathways is the establishment of a block to PC differentiation and high-affinity Ab responses. Unexpectedly, however, the mitochondrial dysfunction and elevated ROS also converged on IFN-receptor signaling and the programming interferon-response functions.

Discussion

We and others have published examples of how metabolic flexibility in B lineage cells may allow limiting or negating the effects of loss-of-function mutations in key steps of intermediary metabolism (54 [↗](#), 55 [↗](#), 97 [↗](#)). These findings raise questions as to forms of intervention that could interfere with the ongoing production of plasma cells secreting pathogenic auto-Ab or, conversely, which pathways might need to collaborate for promoting full-strength Ab responses. The work presented here supports three central insights. First, we have identified conditions under which the anti-NP response is relatively glutaminase-independent, both in the primary phase and upon recall stimulation of memory B cells. Despite this apparent independence, this enzyme central to processing of glutamine and to anaplerosis takes on a vital role if mitochondrial pyruvate import is impeded. A related point is that the findings underscore the fallacy in stating conclusions as if universally applicable, inasmuch as the effect of B cell specific loss-of-function for GLS depended on the nature of the immunization. Thus, the anti-ovalbumin response derived from GLS-depleted B cells was substantially reduced even though the anti-NP responses after multivalent hapten-carrier immunization with NP were not similarly reduced with isolated *Gl*s gene inactivation. Mechanistically, the findings revealed that glutaminase activity promotes a progressive increase in mitochondrial respiration in a manner enhanced by pyruvate import, an alternative pathway for providing substrates to the TCA (Krebs) Cycle. A third major finding is that the efficiency of coupling IFN receptors to the tyrosyl and serine phosphorylation of STAT1 - and IL-21 stimulation to STAT3 - depends on the programming of intermediary metabolism in a manner linked to ROS homeostasis. Taken all together, the findings reveal not only new insights into how intermediary metabolism can modulate differentiation and receptor signaling, but suggest the potential for new interventions in autoimmune diseases driven by interferon effects and auto-Ab.

A self-reinforcing network of increased interferon production and pathological levels and persistence of auto-Ab production is characteristic of several autoimmune conditions of humans. Systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA) are most common and studied, but this pairing also is involved in inflammatory myopathies and primary Sjogren's syndrome (98 [↗](#), 99 [↗](#)). Effectiveness of therapeutic mAb for B cell depletion is best established in RA; unimpressive results in SLE raised concerns that disease might be perpetuated by long-lived PC that secrete auto-Ab, do not need BAFF, and are CD20^{neg}. However, dramatic results of testing CD19-directed chimeric Ag receptor T cells in patients with SLE and myositis showed that targeting B cells reduced clinical disease, IFN levels and those of pathologic auto-Ab [80, reviewed in (81 [↗](#))]. These effects imply that ongoing production of PC from B cells maintains IFN production and disease activity. Cost, production complexity, and risk profile mean that a need to identify other approaches that restrict the ongoing differentiation of B cells into PC remains, despite the

dramatic results in patients with severe disease. The capacity of CB839 to collaborate with UK5099 to suppress both PC differentiation and responsiveness to IFNs suggests that this agent and glutaminase-specific inhibition in general merit further analysis. Since CB839 appeared safe in patients during phase 2 clinical testing (100 [↗](#)), our finding of an interplay between the standard-of-care agent hydroxychloroquine and CB839 in the inhibition of plasma cell development in vitro similarly militates in favor of translational investigation. Which B cells are major sources of the PC that secrete pathological auto-Ab, and the extent to which SLE susceptibility might involve a distinct relationship between intermediary metabolism and PC differentiation, are important issues beyond the scope of the present study. Our data indicate that inhibition of the pathways impacted memory B cells that rapidly differentiated after a secondary challenge, but do not specifically test the atypical MBC that are implicated in SLE pathogenesis (101 [↗](#)).

Another finding of the work bears both on consideration of the intersection of B cell metabolism and function after immunization, and perhaps on the relationship of our findings to SLE. Our data show that although the Ag-specific Ab response elicited by a protein, ovalbumin, was reduced by B cell-specific depletion of GLS in an allergic lung inflammation model of immunization, the anti-NP response was normal. Our finding that the metabolic requirements for Ab responses differed according to Ag or type of immunization raises questions that merit being worked out in future research. A key point is that immunization to elicit anti-NP responses usually is carried out with heavily haptenated carriers, i.e., many NP-groups on each protein molecule. In sharp contrast, a particular epitope in the carrier protein (ovalbumin, KLH, gamma globulin) will be monovalent unless part of an aggregate. Valency and the extent of BCR cross-linking influence signaling by the BCR, which in turn can affect metabolic regulators such as mTORC1 and c-Myc. A related possibility derives from the characteristics of the anti-NP response, which is dominated by a limited Ig heavy chain repertoire (102 [↗](#)). Thus, intrinsic biochemical characteristics of $V_H125:V_L\lambda$ BCR interactions with NP might lead to programming metabolic requirements that differ from what is elicited by the BCR repertoire responding to ovalbumin. The issue of valency has a bearing on the pathological Ab in auto-immunity in that so many auto-Ab of SLE are directed against multivalent Ag such as dsDNA and ribonucleoprotein complexes (3 [↗](#), 5 [↗](#), 103 [↗](#)). Finally, the specific immunization regimen might affect the results. General issues of valency in engagement of BCR at the cell surface remain incompletely clarified despite years of papers, so extensive further studies will be needed to understand better the relationship to a specific step in intermediary metabolism. Nonetheless, a striking finding herein is that a vital requirement for glutaminolysis is cryptic in the anti-NP response until uncovered by reduced import rates of either glucose into cells or pyruvate into mitochondria.

Our findings connect to questions of metabolic flexibility, which is a central issue for the potential translatability or implications of intermediary metabolism in targeting immune responses or auto-immunity. Targeting a particular process is less likely to be efficacious to the extent that GC B cells, or B cells in general, can use alternatives and achieve normal responses or outputs. For instance, although there is evidence that the majority of energy generation by activated and GC B cells involves fatty acid (FA) oxidation, most of it in mitochondria (97 [↗](#), 104 [↗](#)), elimination of a key step in mitochondrial acquisition and use of FAs had no impact on Ab responses or discernible reduction in GC or affinity maturation (97 [↗](#)). Alternatively, restricting cell-intrinsic generation of serine from a glycolytic intermediate represents one potential point of vulnerability that was not compensated by flexibility (58 [↗](#)). Depletion of a major glucose transporter - and inferentially of reduced supply of glucose for some combination of NADPH regeneration from NADP, serine biosynthesis, oxidation, and perhaps other use of glucose - also reduced GC size, Ab responses, and affinity maturation (54 [↗](#), 55 [↗](#)). However, the effect magnitude (reduction to ~1/4 normal) appeared less promising from the standpoint of potential treatment of autoimmunity. Metabolomic data suggested that increased anaplerosis was a compensatory mechanism that might have limited the severity of the defect in ASC generation and Ab responses of GLUT1-depleted B cells (55 [↗](#)). Our findings support this inference in that the defects with *Slc2a1* Δ/Δ or *Mpc2* Δ/Δ B cells were each dramatically amplified by impedance of glutaminolysis. The size of the

differences, the impact on the IgG2(a)c isotype that is particularly important in SLE pathogenesis [reviewed in (105)], and the vital functions of STAT1 induction by IFN receptors in autoimmune models (106, 107) indicate that this synthetic auxotrophy merits translational consideration.

Materials and methods

Mice and immunizations

All animal protocols - reviewed and approved by Vanderbilt University Institutional Animal Care and Use Committee - complied with the National Institutes of Health guidelines for the Care and Use of Experimental Animals. Mice were housed in specified pathogen-free conditions. Both male and female mice, aged 6-10 weeks, were used; sex-specific subgroup analyses did not reveal any significant differences and yielded the same conclusions. To enable tamoxifen-induced, B cell type-specific inactivation of GLS, GLUT1, and/or MPC2 coding potential, *Gls^{ff}* (72), *Slc2a1^{ff}* (35), and *Mpc2^{ff}* mice (60) were crossed with *huCD20-CreER^{T2}* (73) transgenic mice; for confirmatory in vitro work with purified B cells, some experiments used *Rosa26-CreER^{T2}* [as in (36)]. Tamoxifen was administered as reported previously (36, 54). To control for potential Cre toxicity in B cell responses or GC B cells, age-matched *Gls^{+/+}*, *Slc2a1^{+/+}*, and *Mpc2^{+/+}*, *huCD20-CreER^{T2}* mice were co-housed with *huCD20-CreER^{T2}* mice that had conditional alleles [*Gls^{ff}*, *Slc2a1^{ff}*, and/or *Mpc2^{ff}*], alone or in combination, and used as wild-type controls.

Mice were immunized, or immunized and boosted, by one or two intraperitoneal (i.p.) injections of 100 µg of 4-hydroxy-3-nitrophenylacetyl hapten (NP) conjugated to ovalbumin (NP-ova, Biosearch Technologies, Novato, CA), emulsified in 100 µL alum [Imject® alum (Thermo Fisher Scientific, Pittsburgh, PA), as described previously (36, 54, 63)], or after this product was discontinued, Alhydrogel 2% (InvivoGen, San Diego, CA). For mucosal challenges to elicit an anti-carrier Ab response, ovalbumin (50 µg dissolved in 20 µL PBS) was instilled intranasally (i.n.) once daily across seven consecutive days starting three weeks after i.p. initial sensitization of mice with NP-ova. Mice were harvested 12 h after the final inhalation. Alternatively, mice were immunized with sheep red blood cells (SRBC) as described (36). Harvested spleens in some experiments were used to purify GC B cells for RNA-seq (detailed below). To assess proliferation rates of GC B cells in vivo, intravital BrdU incorporation measured by injecting mice with 2 mg BrdU (Sigma-Aldrich, St. Louis, MO) 16 h and 4 h before harvesting spleens from SRBC-immunized mice. Single-cell suspensions, prepared as described above, were stained for surface markers (B220, IgD, GL7, CD38; dump channel markers as specified in the Flow Cytometry section) followed by cell fixation, permeabilization, and staining BrdU-containing DNA using the APC BrdU Flow Kit (B-D Pharmingen, San Jose CA) according to manufacturer's protocol. GC B cells were defined as B220⁺ GL7⁺ IgD^{neg} CD38^{neg}, as control analyses have shown that a CD138⁺ pre-plasma cell population is < 2% of this gate.

Flow cytometry

Fluor-conjugated mAbs were purchased from BD Pharmingen (San Jose, CA), Life Technologies, eBiosciences (San Diego, CA), or Tonbo Biosciences. For detection of GC- and memory-phenotype B cells in the spleens of immunized mice, samples were stained as previously described (22). In brief, 3 x 10⁶ splenocytes were stained with anti-B220, -GL7, -Fas, -IgD, -CD38, NP-APC and a dump channel containing anti-CD11b, -CD11c, -F4/80, -Gr-1, and viability marker (7-AAD or Ghost-Brilliant Violet 510) in 1% BSA and 0.05% sodium azide in PBS. For detection of PC in products of *in vitro* cultures, viable cells (gated via FSC, SSC, and Ghost e450) were stained with fluorophore-conjugated anti-CD138, -B220, -CD19, or -TACI. For flow analyses of mitochondrial and total intracellular ROS, cells (1-3 x 10⁶) were washed in PBS and stained with 40 nM MitoTracker Green, 10 nM TMRE, 5 µM MitoSOX or 1.25 µM H₂DCFDA, respectively, along with Ghost-780 in PBS (20 min at 37°C), then washed again (1% BSA in PBS) and further stained with anti-B220, -CD19, -

CD138, or -TACI. Samples were analyzed using a FACS Canto flow cytometer driven by BD FACS Diva software or as part of preparative flow purification with a FACS Aria flow sorter. Data were processed using FlowJo software (FlowJo LLC, Ashland, OR).

Cell cultures & reagents

B cells were purified from mouse spleens using negative selection with biotinylated anti-CD90 and an iMag system (BD-Pharmingen) or positive selection with anti-mouse B220 nanobeads (Miltyeni Biotec, Auburn CA). To induce plasma cell differentiation, B cells (seeded at 5×10^5 / mL) were cultured with 1 μ g/mL anti-CD40 (Tonbo, San Diego CA), 10 ng/mL BAFF (AdipoGen, San Diego, CA), 10 ng/mL IL-4 (Peprotech, Rocky Hill, NJ), 5 ng/mL IL-5 (Peprotech), and, if derived from inducible gene deletion models, 50 nM 4-hydroxy-tamoxifen (4-OHT) (Sigma-Aldrich, St. Louis MO) and re-fed at day 3. When cultured 2 d or less, cells were in the same conditions after plating at 2×10^6 /mL. Alternatively, B cells were activated with both Fab2' anti-mouse IgM (Southern Biotech, Birmingham, AL) as well as 1 μ g/mL anti-CD40 and cultured as for anti-CD40-activated cells. Iutamine supplementation analyses, with or without added DMK (5 mM) were performed using glutamine-free RPMI (Thermo Fisher, Waltham MA) supplemented with 10% dialyzed Gibco FBS (Thermo-Fisher), 100 U/mL penicillin (Invitrogen, Waltham MA), 100 μ g/mL streptomycin (Invitrogen), non-essential amino acids (NEAA, Invitrogen), 10 mM HEPES (Invitrogen), 0.1 mM 2-mercaptoethanol (Sigma). To test how inhibitors affected in vitro proliferation, cytokine receptor signaling, or plasma cell differentiation, cultures were supplemented with 1 μ M CB839, 10 μ M UK5099, and/or 3 μ M hydroxychloroquine (all from Cayman Chemical Co, Ann Arbor MI), individually or in combinations, and 1 mM DMK was used to supplement some cultures subjected to glutaminolysis inhibition by CB839. To analyze proliferation in vitro, B cells (2×10^6 , purified as above) were labeled with 5 μ M CellTrace Violet (CTV) (Invitrogen, Waltham, MA) and then activated and cultured as above. To test the impact of ROS and electron transport chain (ETC) inhibition on activation of STAT transcription factors, cultures of B cells activated with anti-CD40, BAFF, IL-4, and IL-5 were supplemented with 0.2 mM H_2O_2 (Sigma-Aldrich), 2 μ M menadione (Sigma-Aldrich), 2 mM metformin (Sigma-Aldrich), or 1 μ M rotenone (Calbiochem, San Diego CA).

ELISA and ELISpot

Relative NP-specific Ab concentrations in sera, and secreted Ab in culture supernatants, were measured using capture ELISA as described previously (36, 54, 63). All- or high-affinity hapten-specific Ab were measured using plates coated with NP₂₀-BSA or NP₂-PSA (Biosearch Technologies), respectively. Ovalbumin-specific antibodies were measured using serial dilutions of sera incubated on ovalbumin (1 μ g/well)-coated 96-well plates (16 h at 4° C). Total secreted Ab were captured using an anti-Ig(H+L) reagent. Captured Ab retained after rinsing were detected in isotype-specific manner using HRP-conjugated anti-IgM, -IgG1, IgG2c, or -IgE incubation and colorimetric development with Ultra TMB Substrate (Thermo Fisher Scientific).

The frequencies of antibody secreting cells (ASCs) in the bone marrow, lung, or spleen after immunization were measured as described previously (54, 63). Briefly, high protein binding plates (Corning Life Science, Corning NY) were coated with NP₂₀-BSA (for all-affinity ASCs) or NP₂-PSA (for high-affinity ASCs) (Biosearch Technologies), then seeded with splenocyte suspensions [$0.5 - 1 \times 10^6$ cells/well (IgM), $1 - 2 \times 10^6$ cells/well (IgG1), and 2×10^6 cells/well (IgG2c)], cultured in medium with 10% FBS (16 hr), and probed with biotinylated anti-IgM, anti-IgG1, or anti-IgG2c antibodies (Southern Biotechnologies, Birmingham AL). For the detection of IgM or IgG1 ASC in day 5 *in vitro* cultures, high protein binding plates were coated with anti-Ig(H+L) (Southern Biotechnologies), seeded with suspensions of cultured cells [20 hr, with 100, 200, and 500 cells per well (IgM) or 1000, 2000, and 5000 cells/well (IgG1)], then probed with biotinylated anti-IgM or -IgG1 antibodies (Southern Biotechnologies, Birmingham AL). ASC numbers and spot sizes were quantified using an ImmunoSpot Analyzer (Cellular Technology, Shaker Heights, OH) and the densities based on dilutions with readily resolved spots.

Nucleic acid analyses

RNA was isolated from sorted B cells or in vitro-generated B lymphoblasts using TRIzol reagent following the manufacturer's instructions (Life Technologies, Carlsbad, CA). RNA from cultured B cells purified from TriZol and processed for bulk RNA-seq by the VANTAGE Core of the Vanderbilts. To assess the extent of the loss-of-function in B cell populations, gene expression was analyzed using quantitative real-time RT-PCR (qRT²-PCR). After cDNA synthesis by reverse transcription with an AMV Reverse Transcriptase kit (Promega, Madison, WI), amplifications were performed using SYBR green Power UP PCR master-mix (ThermoFisher Scientific) and primer pairs specific for *Gls* (Forward 5'-GGGAATTCACCTTTGTCACGA -3'; Reverse 5'-GACTTCACCC TTTGATCACC -3'), *Mpc2* (Forward 5'-TGTTGCTGCCAAGAAATTG-3'; Reverse 5'-AGTGGACTGAGCTGTGCTGA -3'), and b-actin (Forward 5'-GGCACCACAC CTTCTACAATG-3'; Reverse 5'-GGGGTGTGAAGGTCTCAAAC-3'). The relative level of gene expression was calculated using the 2-DDCT method after each sample was normalized to b-actin. Libraries for low-input bulk RNA-seq analyses with flow-purified GC B cells were generated and sequenced essentially as described (108 [DOI](#)) and sequenced at the Emory Integrated Genomics Core. In brief, For each sample 1,000 cells were sorted directly into RLT buffer (79216; Qiagen) containing 1% (v/v) 2-mercaptoethanol. RNA was isolated using Zymo Quick-RNA MicroPrep Kit (11-328M; Zymo Research). Synthesis of cDNA was performed using SMART-Seq v4 Ultra Low Input RNA Kit (634894; Takara Bio) kit. Final libraries were generated using 200 pg of cDNA as input for the NexteraXT kit (Illumina, FC-131-1024) with 12 cycles of PCR amplification. Final RNA-seq libraries were quantitated by QuBit (Life Technologies, Q33231), size distributions determined by bioanalyzer (Agilent 2100), pooled at equimolar ratios, and sequenced at the Emory Genomics Core on a NovaSeq6000 using a PE100 run. The RNA-seq samples of a biologically independent replication set of flow-purified GC B cells from immunized mice representing the four different genotypes were sequenced at Novogene on a NovaSeq X Plus using a PE150 runBulk RNA-seq. RNA from cultured B cells was performed by similar methods at the Vanderbilts' VANTAGE core.

To analyze sequencer outputs of each core, Fastq files were assessed for quality using FastQC. Adapters were trimmed using TrimGalore, followed by FastQC to validate adapter removal. hisat2 was used to align fastq files to the mm10 genome downloaded from UCSC genome browser. Qualimap was used to analyze the aligned sequences for quality, Subread's featureCounts was used to count and normalize sequence counts in groups being compared. featureCount outputs were then used for DESeq2 analysis to identify fold change and significance of observed differences in gene expression. Normalized RNA-Seq counts were analyzed using GSEA 4.3.2 (Broad Institute) and the Hallmark gene sets collection from the Molecular Signatures Database (MSigDB) to identify changes in biological pathways. Differentially expressed genes identified by DESeq2 were analyzed using the MyGeneset tool from the Immunological Genome Project (Immgen) to identify B cell subsets associated with genes effected by intervention in metabolic pathways. The web-based gene set analysis toolkit (WebGestalt) was used for gene ontology over-representation analysis of biological processes using the protein coding genome as the reference set. Differentially expressed genes identified by DESeq2 were analyzed with WebGestalt to identify biological processes implicated by the change in gene expression.

Metabolic analyses

For proton nuclear magnetic resonance spectroscopy (¹H-NMRS) quantitation of metabolites in solution, plasma and lymph node interstitial fluids were collected and processed as previously reported (67 [DOI](#), 68 [DOI](#)). In brief, a total of 50 µL deuterated water (D₂O) and 50 µL of 0.75% sodium 3-trimethylsilyl-2,2,3,3-tetradeuteriopropionate (TSP) in D₂O were added to 500 µL of diluted biological fluids and transferred to 5-mm NMR tubes (Wilmad-LabGlass, Kingsport, TN). ¹H-NMRS spectra were acquired on an Avance III 600 MHz spectrometer equipped with a Triple Resonance CryoProbe (TCI) (Bruker) at 298 K with 7500-Hz spectral width, 32,768 time domain points, 32 scans, and a relaxation delay of 2.7 seconds. The water resonance was suppressed by a gated

irradiation centered on the water frequency. The spectra were phased, baseline corrected, and referenced to TSP using Chenomx NMR Suite. Spectral assignments were based on literature values.

Oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) were measured using a Seahorse Bioscience XFe96 extracellular flux analyzer (Agilent Technologies, Santa Clara CA). B cells (2.5×10^5) activated for 2 days with anti-CD40, BAFF, IL-4, and IL-5 were seeded per well of a Cell-Tak (5 $\mu\text{g/mL}$; Corning) coated plate. Glycolytic and mitochondrial stress tests were performed as previously described (54, 63). Basal respiration, maximum respiration, and ATP production were calculated using formulas derived from the software accompanying the Agilent Seahorse platform. Steady-state ATP concentrations in activated B cells were measured using the Promega Mitochondrial ToxGlo™ Assay (Promega, Madison WI), following the manufacturer's protocol. B cells activated and cultured 2 d in anti-CD40, BAFF, IL-4, and IL-5 after negative selection of splenocytes with anti-Thy1.2, in the presence or absence of metabolic inhibitors (CB839, UK5099, or both). Equal numbers of viable (Trypan Blue^{neg}), washed twice with warm, sterile phosphate-buffered saline, were resuspended in glucose-supplemented RPMI1640, with their designated drug treatment added, and cultured (10^5 B cells in 100 μL) 90 min at 37°C. Cytotoxicity Reagent (20 μL 5x) was added to each well followed (30 min) by quantifying fluorescence (520-520 nm Em; 485nmEx/). 100 μL of ATP Detection Reagent was then added to each well followed by orbital shaking (500rpm for 5 min). Luminescence was then measured to quantify ATP present in the B cells.

For metabolomic analyses, frozen (-70 C) cell pellets were extracted in plastic vials with 40 μL of UPLC grade isopropanol, sonicated (3 cycles, 10 seconds each), and clarified by centrifugation (20 min). Total protein content was measured using Pierce BCA Protein Assay kit (Fisher-Thermo Scientific). Supernatants were analyzed using the Biocrates MxP Quant 500 XL targeted metabolomic kit (Biocrates Life Sciences AG, Innsbruck, Austria), potentially quantitating 106 small molecules and free fatty acids in chromatography mode and 913 complex lipids in flow-injection mode (FIA-MS/MS).

Metabolite identification was based on triple quadrupole ultra-high-performance liquid chromatography tandem mass spectrometry (UHPLC-MS/MS) using a Shimadzu Nexera chromatography platform (Shimadzu Corporation, Kyoto, Japan) coupled to Sciex QTRAP 7500 mass spectrometer (AB Sciex LLC, Framingham, Massachusetts, USA). Metabolic indicators (474) were then calculated from sums or ratios of relevant metabolites according to Biocrates MetaboINDICATOR formulas (110). These indicators can be regarded as physiologically relevant measures and are statistically analyzed separately from metabolites. In addition, indicators denoted as "X synthesis" were computed as a ratio of metabolite X and its main precursors as a reflection of the inferred conversion ratio. Using solvents of LC/UHPLC-MS grade, cleared extracts were transferred onto a kit plate with pre-injected internal standards, dried, derivatized with 5% phenylisothiocyanate in pyridine, ethanol and water (1:1:1 v/v/v), and subsequent extraction with 5 mM ammonium acetate in methanol. UHPLC was performed with 0.2% formic acid in acetonitrile (organic mobile phase) and 0.2% formic acid in water (inorganic mobile phase). Flow-injection analysis was performed with methanol and Biocrates MxP Quant 500 XL additive. Sample handling, randomized with stratification prior to processing to avoid potential biases, was done on dry ice to avoid multiple freeze-thaw cycles. Plates included isopropanol blanks to calculate limits of detection, repeats of a kit quality control sample to calculate concentrations and monitor the coefficient of variation, and kit calibrators for seven-point calibrations of certain compounds. Analytes (metabolic indicators) were calculated according to Biocrates MetaboINDICATOR formulas (110). Ratios with zeros in denominator were treated as missing values that were not considered in the analysis. As additional data transformations, Box-Cox transformation with R package *car* (111), and Tukey's fencing (112) were applied for metabolomic data. Furthermore, the values were standardized with respect to the DMSO group to facilitate comparison of regression coefficients in the statistical analysis.

Measurement of cytokine-induced STAT phosphorylation

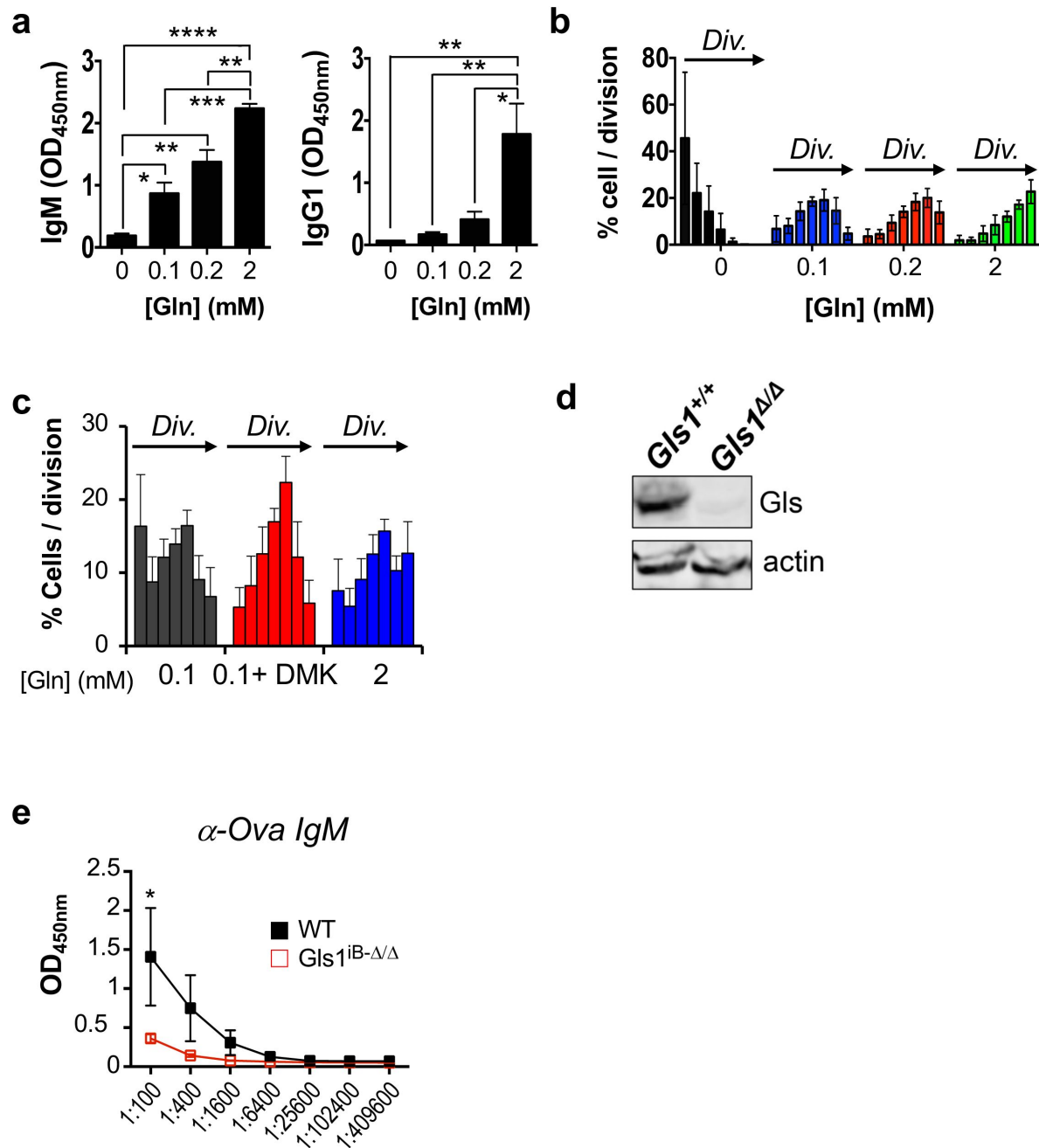
To measure the levels of total and phosphorylated proteins, B cells were stimulated with anti-CD40 and BAFF, cultured in the presence or absence of inhibitors as detailed in **Figure 6** [↗](#) and its Legend, and stimulated (25 min) with IFN β , IFN γ , or IL-21. Total cell lysates were prepared with modified RIPA lysis buffer (50 mM Tris $\times$ Cl (pH8.0), 150 mM NaCl, 1% NP-40, 0.1% SDS, 1 mM EGTA, 1 mM Na₂VO₄, 1 mM NaF, and protease inhibitor cocktail). Proteins resolved by SDS-PAGE were transferred onto ImmobilonTM Nylon membranes (Millipore) that were then incubated with anti-phospho-STAT1-Y701 Ab, anti-phospho-STAT1-S727 Ab, anti-phospho-STAT3-Y705 Ab, anti-GLS1 Ab, or anti-cyclophilin Ab followed by fluorophore-conjugated, species-specific secondary anti-Ig Abs (Rockland Immunochemicals, and LI-COR). Proteins were visualized and quantitated by Odyssey infrared imaging scanner (LI-COR).

Statistical analyses

Data are reported based on results replicated in at least three biologically independent performances. The primary analyses were conducted on pooled data points from independent samples and replicate experiments, *t* testing with post-test validation of its suitability. Tests used in particular settings are generally indicated in the Figure Legends. In experiments with individual analyses and comparisons of limited numbers of samples, *t*-tests or non-parametric formulae were applied as appropriate, based on modeling of the similarity of variances between data from the two conditions being compared, and of the estimated fit to Gaussian distributions; Mann-Whitney testing was used if statistical analysis indicated that the two samples sets probably had unequal variance. Data are displayed as mean ($\pm$ SEM); conventionally, a difference is considered "statistically significant" when the *p* value of for the null hypothesis of a comparison was <0.05 . In addition to application of this method to individual features in the metabolomic analyses, a series of multivariable linear regression models were employed, each corresponding to a specific analyte, to analyze differences in metabolomic data. In these models, the values of the analytes served as the dependent variables, while individual non-DMSO treatment groups were included as key independent variables, i.e. considering the DMSO group as the reference.

Supplemental information

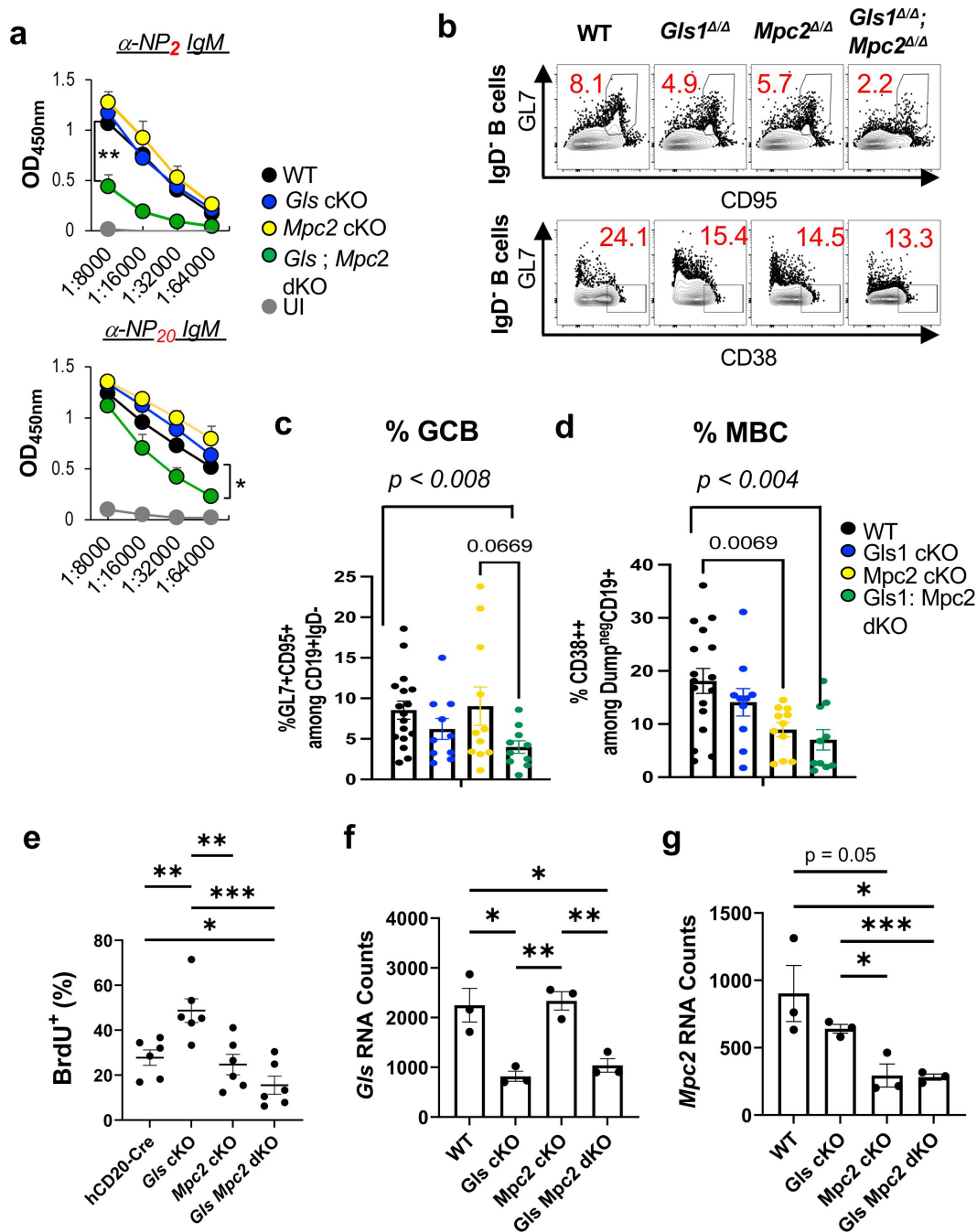
Supplemental figures



Supplemental Figure 1

Glutamine and glutaminolysis promote an antibody response to ovalbumin.

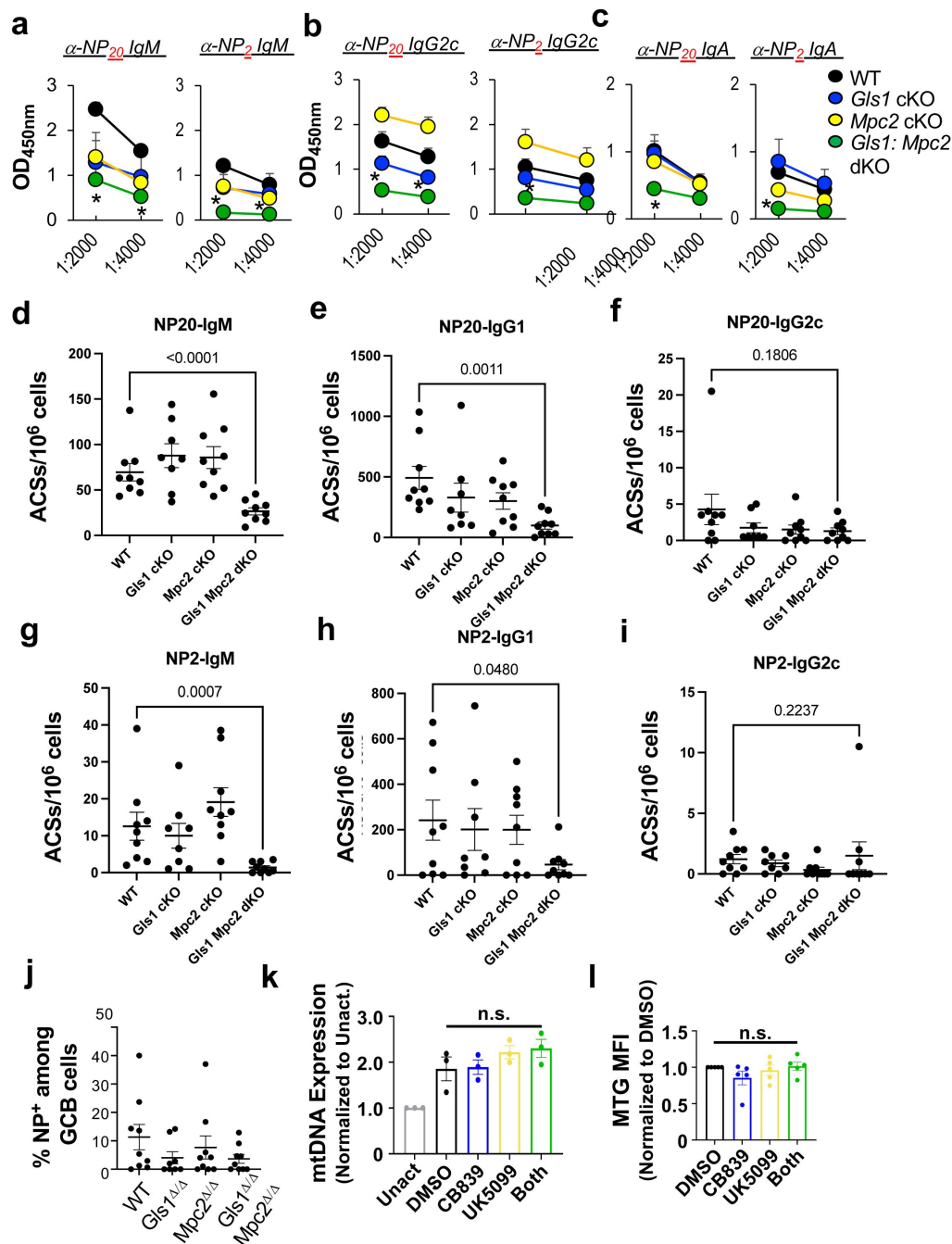
Additional data from or relating to experiments in [Figure 1](#). (a) Using a dilution in the linear range for the samples, relative concentrations of IgM (left panel) and IgG1 (right panel) in the supernatants of cultures of [Fig. 1b](#) were measured by ELISA, using titrations of re-added glutamine concentrations as indicated. Shown are the mean ($\pm$ SEM) values from analysis of supernatants in three biologically independent experiments, with indications of P values as in the main figure. (b) Distributions of division counts under conditions of lower extracellular glutamine in experiments of [Fig. 1b](#). Shown are the percentages of live cells in the gate for each peak of 2-fold CTV partitioning when grown under the indicated conditions. (c) Distributions of division counts under conditions of lower extracellular glutamine, with or without addition of dimethylketoglutarate (DMK) as indicated, in experiments of [Fig. 1c](#). (d) Relative levels of ovalbumin-specific IgM in experiments of [Fig. 1d-i](#), showing mean ($\pm$ SEM) absorbances in ELISA data across serial dilutions, as in [Fig. 1i](#).



Supplemental Figure 2

Synthetic auxotrophy of glutaminolysis in primary anti-NP Ab responses.

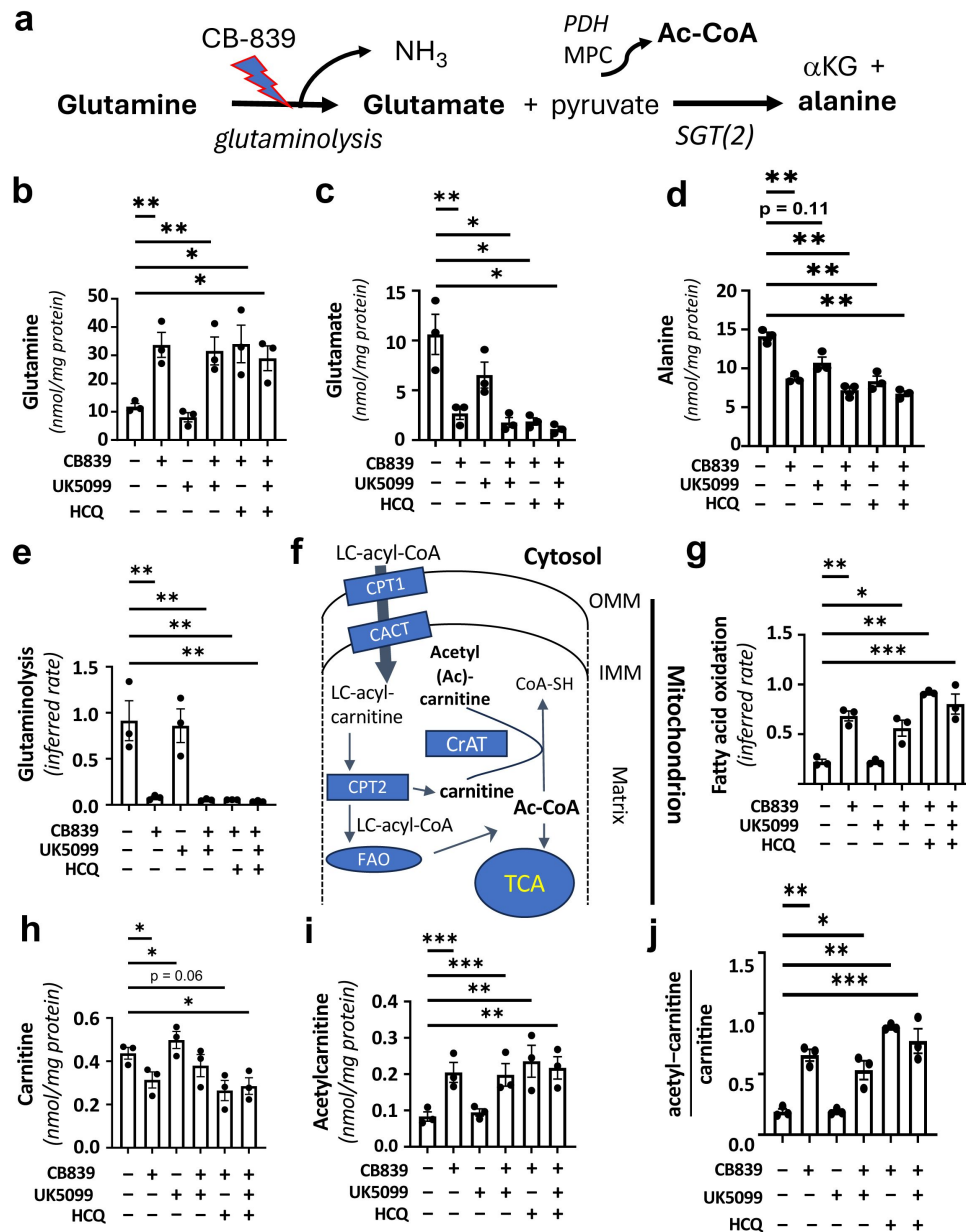
(a-d) Mice of indicated genotypes were treated with tamoxifen and immunized as in [Figure 3a](#). (a) All- and high-affinity (lower and upper, respectively) IgM anti-NP Ab in sera at time of harvest (same samples & display as in [Fig 3b, c](#)). (b-d) Spleens were harvested from NP-OVA boosted mice and the frequencies of GL7⁺ CD95⁺ GCB cells and GL7⁺ CD38⁺ memory B cells (MBCs) were measured by flow cytometry. (b) Gating for measurements of prevalence in flow cytometric data, showing representative flow plots for GCB cells and MBCs in IgD⁺ B cell gated cells. (c, d) Shown are mean ($\pm$ SEM) frequencies of (c) GCB cells and (d) MBCs from five independent replicate experiments. (e) Intravital incorporation of BrdU by GCB cells after priming immunization, tamoxifen injection, and boost immunization as described in the Methods. (f, g) RNA-Seq normalized read counts of inducible gene knockout targets in flow sorted SRBC stimulated GCB cells. Related information is presented in [Supplemental Fig. 8a, b](#).



Supplemental Figure 3

Glutaminase and mitochondrial pyruvate channel promote response of reactivated memory B cells.

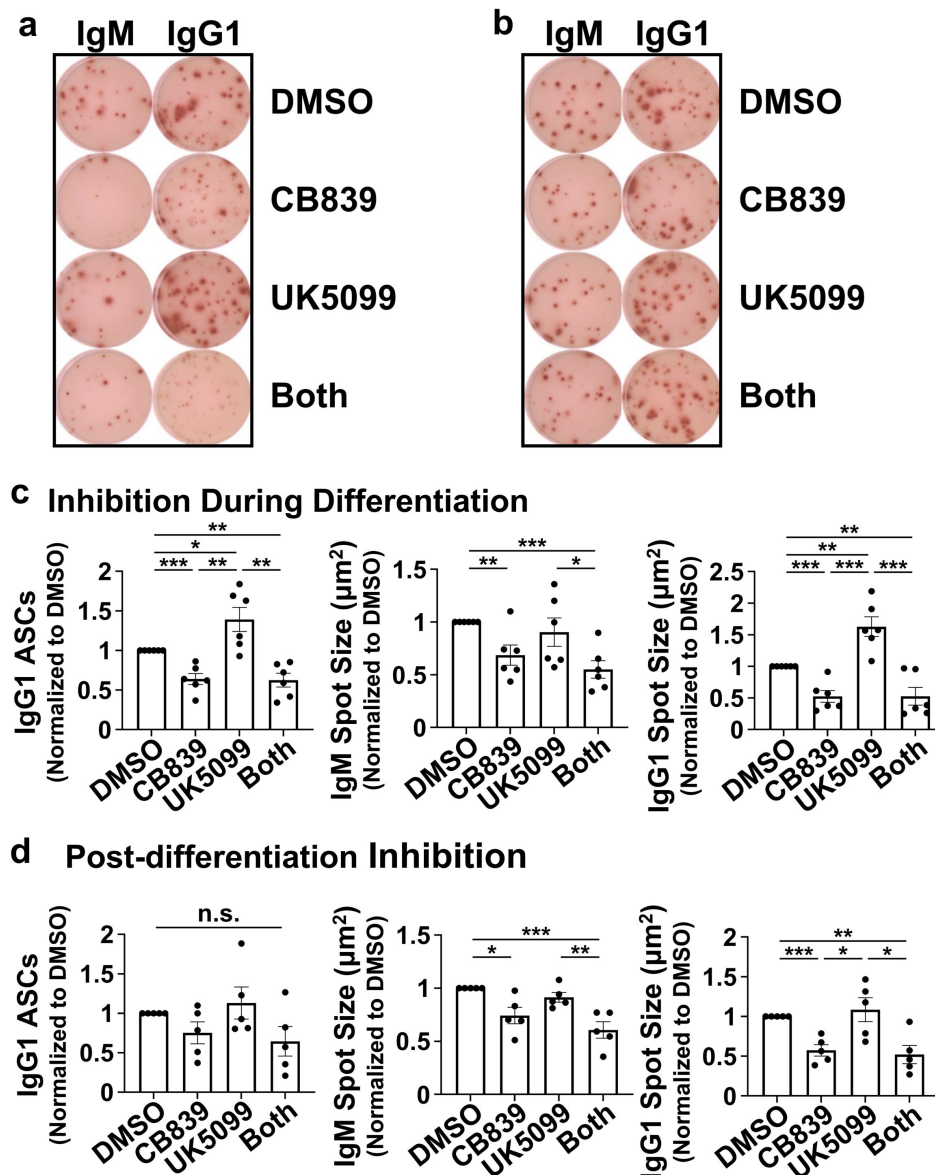
Mice of indicated genotypes were immunized with NP-OVA, injected with tamoxifen and boosted with NP-OVA as in [Fig 3h](#). (a-c) Shown are serologies of the all- and high-affinity NP-specific IgM (a), IgG2c (b) and IgA (c) antibodies from 3 independent experiments. (d-f) Shown are mean ($\pm$ SEM) numbers of NP-specific IgM-, IgG1-, and IgA-secreting cells in the splenocytes from immunized mice. (g-i) Frequencies of NP-specific ASCs in the spleens from immunized mice with indicated genotypes were analyzed by ELISpot assays. Shown are mean ($\pm$ SEM) frequencies of high-affinity anti-NP (g) IgM, (h) IgG1, and (i) IgG2c. (j) NP-specific GC B cells were analyzed in the spleen from immunized mice with indicated genotypes. Shown are mean ($\pm$ SEM) frequencies of NP⁺ cells in GL7⁺ CD95⁺ gated B cells from 3 independent replicate experiments. (k, l) Connecting to [Fig 5](#), mitochondrial (mt)DNA content (k) and MitoTracker Green (MTG) labeling of B lymphoblasts generated as for [Fig 5](#) were measured by qPCR or flow cytometry, respectively.



Supplemental Figure 4

Impact of CB-839 and UK-5099 on glutaminolysis and mitochondrial metabolism.

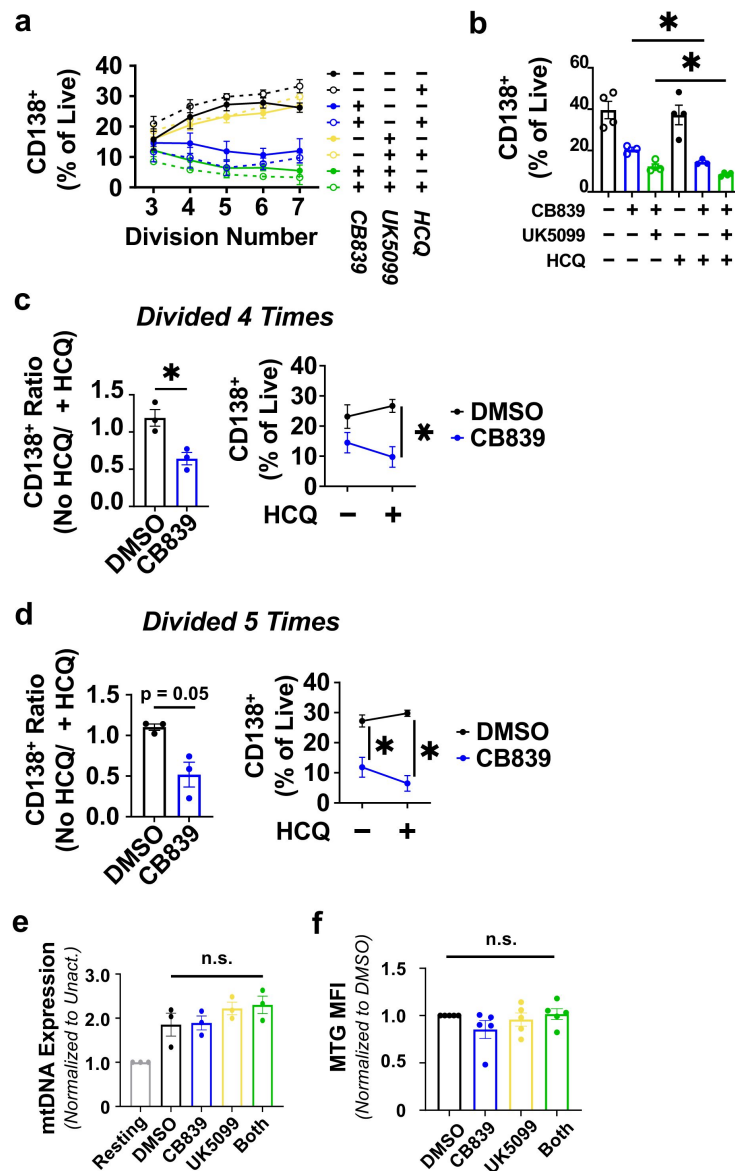
(a) A schematic illustration of glutaminolysis, its inhibition by CB-839, and GPT2-(mitochondrial alanine aminotransferase, also termed glutamate pyruvate transaminase) catalyzed conversion of glutamate and pyruvate for generation of alanine. PDH, pyruvate dehydrogenase; MPC, mitochondrial pyruvate channel. (b-j) Selected metabolic perturbations identified by metabolomic analysis of activated B cells. For each of three biologically and temporally independent replicate experiments, B cell pools were purified from several mouse spleens, then activated and cultured 2 d as in Fig 5, or under the indicated conditions [or with Fab2' anti-IgM (1 µg/mL) added, yielding similar results (not shown)]. Shown are results for (b) glutamine, (c) glutamate, (d) alanine, along with results of calculating inferred glutaminase activities (e). (f) Schematic illustrating mitochondrial import of fatty acids via L-carnitine and acyl-carnitine intermediaries, along with use of the acetyl (Ac)-CoA for either entry into the Krebs (TCA) cycle or generation of acetyl (Ac)-carnitine. LC, long-chain; coA, coenzyme A; CPT, carnitine palmitoyl transferase, FAO, fatty acid oxidation; acetyl, Ac; TCA, tricarboxylic acid (Krebs cycle); CACT, carnitine-acylcarnitine transferase; CrAT, carnitine O-acetyltransferase. (g) Computationally inferred activity of fatty acid oxidation (FAO) derived from the metabolomic data. (h) L-carnitine and (i) acetyl-carnitine concentrations in the activated B cells, and the ratio calculated for each sample (j).



Supplemental Figure 5

Glutaminolysis enhances rates of Ab secretion by PC in addition to promoting their development.

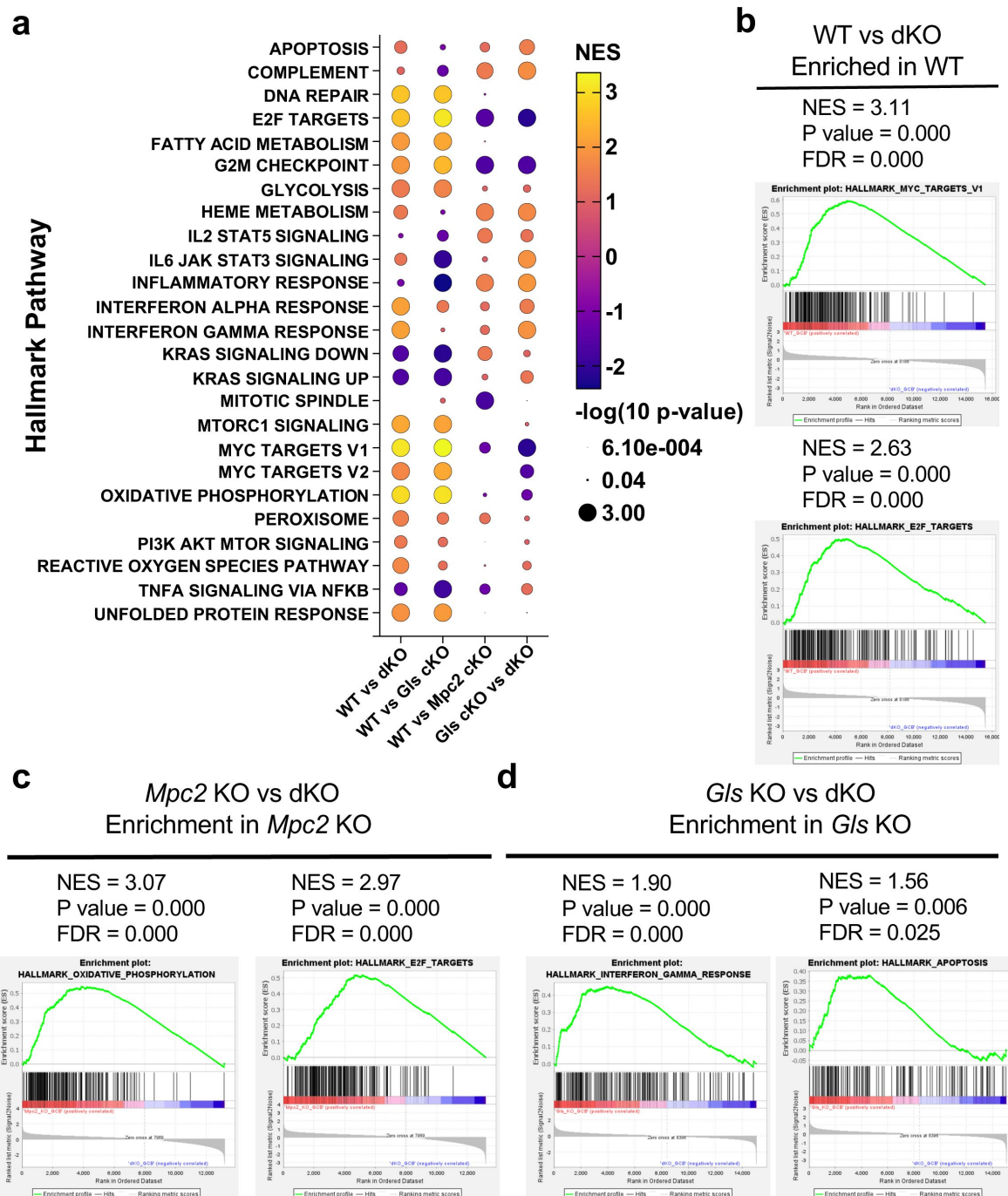
Using conditions akin to **Fig. 4a-e**, purified B cells were activated and cultured 4 d in the presence or absence of the indicated inhibitors. Equal number of viable cells were replated for ELISpot assays after rinsing and counting, with cultures in the wells performed in the absence (panels a, c) or presence (b, d) of the indicated inhibitors. (a, b) Photographs of spots in wells of the indicated cultures, scoring the frequencies as well as spot sizes of cells secreting IgM or IgG1 as indicated. Shown are single wells from one representative experiment, representative of the technical duplicates and of the five biological replicate experiments. (a) B cells were cultured 4 d in the presence of CB839 or UK5099 prior to plating culturing in inhibitor-free medium for the ELISpots. (b) B cells were cultured 4 d in inhibitor-free medium, followed by addition of the indicated compounds after plating in the ELISpot wells and overnight cultures. (c, d) Mean ($\pm$ SEM) data from all five biologically independent replicate experiments quantitating the relative frequencies of ASCs and sizes of the spots (surrogates for amount of Ab secreted during the overnight culture) are shown. Left, middle, and right panels show relative frequencies of ASCs secreting IgG1, mean spot sizes after detection of IgM, and mean IgG1 spot sizes, respectively. For each experiment (a common pool of purified B cells activated and cultured in parallel with inhibitor(s) or vehicle alone), the ASC numbers and average spot sizes for inhibitor-treated cultures were normalized to those measured for the vehicle (DMSO) control. (c) Inhibitors were present during 4 d cultures, as in (a). (d) Inhibitors were added only after plating in ELISpot wells for overnight assays of secretion after a pool of activated B cells was aliquoted after culture 4 d without inhibitor present,



Supplemental Figure 6

Combinatorial reduction in PC differentiation k hydroxychloroquine and glutaminolysis inhibition includes a division-independent mechanism.

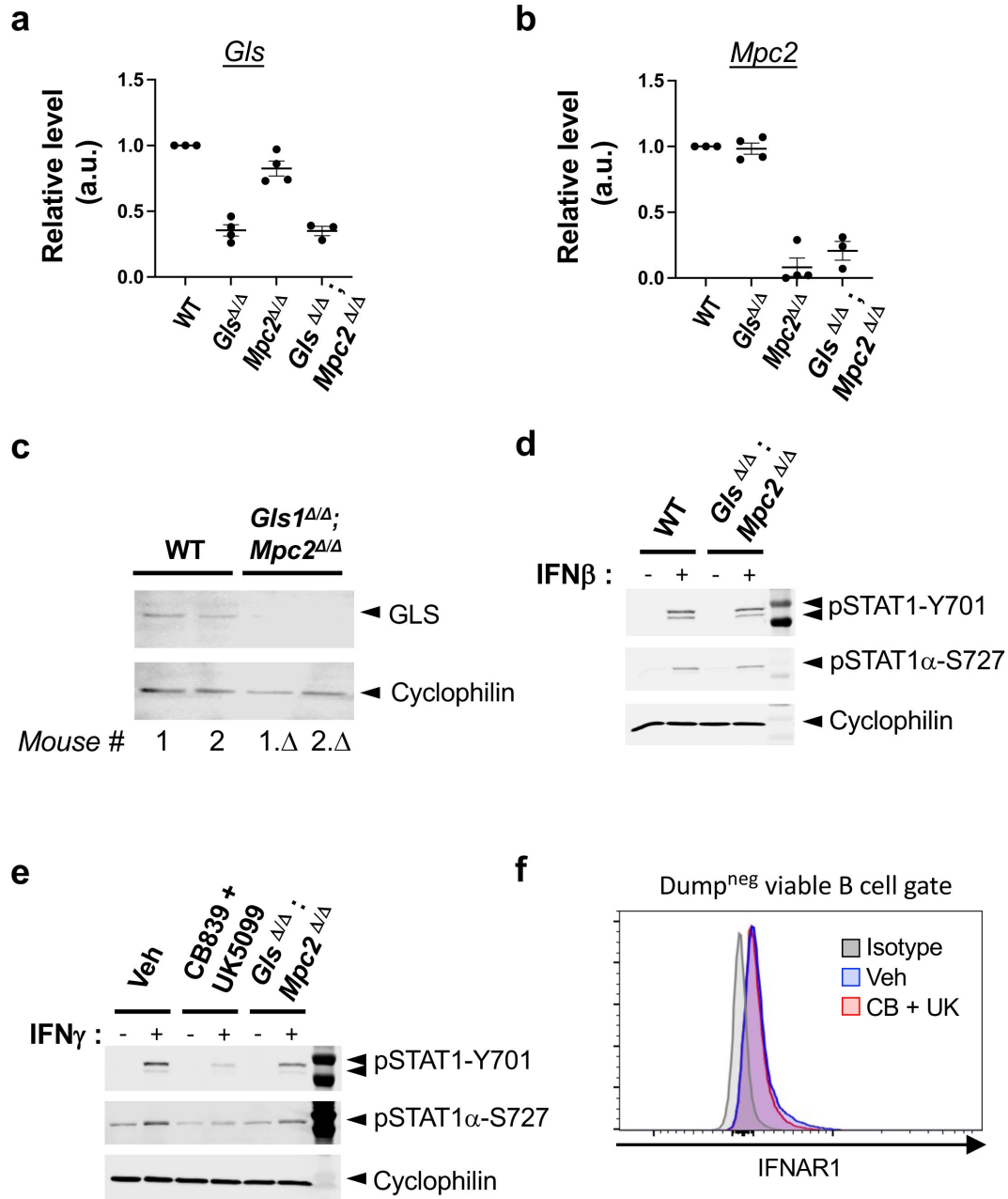
(a) Plasma cell differentiation cultures of purified B cells, labelled with CTV and activated in the presence of the indicated combinations of drugs (or DMSO vehicle), were performed as in Fig. 4 and analyzed by flow cytometry. (a) Mean ($\pm$ SEM) %CD138⁺ cells (day 5) at levels of CTV fluorescence representing divisions 3 through 7 are plotted separately for each condition shown in the key. Open symbols, HCQ added; filled symbols - no HCQ. Line colors are coded as in Fig. 4, 5. Mean results derived from three biologically independent experiments. (b) Quantitative data on frequencies of plasma cells after independent cultures of purified B cells were performed and analyzed as in Fig. 4 (no CTV labeling), in the indicated combinations of drugs. Dots denote individual values for four independent B cell pools and cultures, with bars representing the mean % CD138⁺. (c, d) HCQ effect on PC development contingent on inhibition of glutaminolysis. Using data from (a), the % CD138⁺ for each indicated condition in each independent experiment was measured in the CTV peaks representing viable cells that divided four (c) and five (d) times. In each case, a ratio of control to HCQ-treated value was calculated for the condition (DMSO or CB-839 present) (left graph) and the actual % CD138⁺ with and without HCQ (right graph). (e) qPCR measurement of mtDNA after activation and culture (2 d) as in Fig. 5c. (f) Cellular mitochondrial content determined by MitoTracker Green (MTG) labelling quantified by flow cytometry. Shown are MFI values from each independent experiment after activation and culture (2 d) as in Fig. 5c, then normalized to DMSO-treated condition in each experiment.



Supplemental Figure 7

Altered gene expression of metabolically reprogrammed B cells

Additional data relating to the analyses of RNA-seq results with flow-purified GC B cells (controls versus those with disruption of *Gls*, *Mpc2*, or both; Fig. 4j-l; Fig. 6a-c). (a) In the bubble plot summarizing the results of GSEA using the RNA-seq data, the heat-mapped color coding of each circle denoted the normalized enrichment score on the scale to the right, while the size of each circle indicates the adjusted P value. (b-d) Selected GSEA plots illustrative of the changes in transcriptional programs of GC B cells with altered metabolism due to post-maturation disruption of the genes *Gls*, *Mpc2*, or both, as summarized in (a). (b) Enrichment of Myc- and E2F-upregulated mRNA in WT samples as compared to *Gls* Δ/Δ , *Mpc2* Δ/Δ . (c) GLS-dependent increases in expression of RNA of the oxidative phosphorylation and E2F pathways, with GSEA comparing *Mpc2* Δ/Δ to *Gls* Δ/Δ , *Mpc2* Δ/Δ GC B cells shown. (d) MPC-dependent increases in expression of RNA of the IFN- γ response and apoptosis program gene sets compared for *Gls* Δ/Δ versus *Gls* Δ/Δ , *Mpc2* Δ/Δ GC B cells shown.



Supplemental Figure 8

Normal IFN-R expression yet decreased P-STAT1 in metabolically reprogrammed B cells

(a, b) Deletion efficiency of *Gls1* and *Mpc2* in vivo. CD19⁺ IgD⁺ naïve B cells were flow-purified from spleens of tamoxifen-injected and SRBC-immunized hu-CD20-CreER^{T2} mice (*Gls1*^{f/f}, *Mpc2*^{f/f}, or wild-type). Shown are the levels of *Gls*- (a) and *Mpc2*-encoded RNA (b) in the cells with the indicated genotypes relative to WT control after normalized to β-actin as an internal control. (c) Immunoblot analysis of whole cell extracts of B cells of the indicated genotypes, purified from tamoxifen-treated mice (two of each genotype probed with anti-GLS and anti-cyclophilin B. (e, f) Active metabolism via GLS1 and MPC2 promotes interferon activation of STAT1. WT or *Gls1*^{Δ/Δ}; *Mpc2*^{Δ/Δ} B cells were activated with anti-CD40 and BAFF for 2 days followed by IFN-β (c) or IFN-γ (d) treatment for 15 min. Shown are the representative western blot images from more than three independent experiments. (f) Cell surface IFNAR1 signal on B cells activated and cultured as in Fig 6 was determined by flow cytometry. Shown is a representative result from replicate experiments (n = 3) analyzing IFNAR1 expression on viable B cells.

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Additional information

Abbreviations used

- 2-DG: 2-deoxyglucose
- 4OHT: 4-hydroxytamoxifen
- αKG: α-ketoglutarate
- Ab: antibody
- Ag: antigen
- ASC: Ab-secreting cell
- BAFF: B-cell activating factor
- 2-deoxyglucose
- BCR: B cell Ag receptor
- BrdU: bromodeoxyuridine
- BSA: bovine serum albumin
- WT: wildtype
- KO: knockout
- MZB: marginal zone B cell
- PC: plasma cell
- MBC: memory B cell
- Ig: immunoglobulin
- CD: cluster of differentiation
- CoA: coenzyme A
- CTV: CellTrace Violet
- DMK: dimethyl-ketoglutarate
- DMSO: dimethylsulfoxide
- ECAR: extracellular acidification rate
- ELISA: enzyme-linked immunosorbent assay
- ELISpot: enzyme-linked immunosorbent spot
- ETC: electron transport chain
- FAO: fatty acid oxidation
- FCCP: carbonyl cyanide 4-phenylhydrazone
- GC: germinal center
- GLS: glutaminase
- GSEA: gene set enrichment analyses

- H2DCFDA: dichlorodihydrofluorescein diacetate
- HIF: hypoxia-inducible factor
- HCQ: hydroxychloroquine
- IFN: interferon
- IL-: interleukin-
- Jak: Janus kinase
- MPC: mitochondrial pyruvate channel
- NP-OVA: 3-Nitrophenylacetyl (NP)-ovalbumin (OVA)
- OCR: oxygen consumption rate
- PDH: pyruvate dehydrogenase
- PSA: porcine serum albumin
- ROS: reactive oxygen species
- mtROS: mitochondrial ROS
- SDS-PAGE: sodium dodecyl-sulfate polyacrylamide gel electrophoresis
- SEM: standard error of means
- SRBC: sheep red blood corpuscule
- STAT: signal transducer and activator of transcription
- TCA cycle: tricarboxylic acid cycle
- TLR: Toll-like receptor
- TMRE: tetramethylrhodamine ester
- Tmx: tamoxifen

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Vanderbilt University Medical Center (007)

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Reviewer #1 (Public review):

Summary:

In this manuscript, Cho et al. present a comprehensive and multidimensional analysis of glutamine metabolism in the regulation of B cell differentiation and function during immune responses. They further demonstrate how glutamine metabolism interacts with glucose uptake and utilization to modulate key intracellular processes. The manuscript is clearly written, and the experimental approaches are informative and well-executed. The authors provide a detailed mechanistic understanding through the use of both in vivo and in vitro models. The conclusions are well supported by the data, and the findings are novel and impactful. I have only a few, mostly minor, concerns related to data presentation and the rationale for certain experimental choices.

Detailed Comments:

(1) In Figure 1b, it is unclear whether total B cells or follicular B cells were used in the assay. Additionally, the in vitro class-switch recombination and plasma cell differentiation experiments were conducted without BCR stimulation, which makes the system appear overly artificial and limits physiological relevance. Although the effects of glutamine concentration on the measured parameters are evident, the results cannot be confidently interpreted as true plasma cell generation or IgG1 class switching under these conditions. The authors should moderate these claims or provide stronger justification for the chosen differentiation strategy. Incorporating a parallel assay with anti-BCR stimulation would improve the rigor and interpretability of these findings.

(2) In Figure 1c, the DMK alone condition is not presented. This hinders readers' ability to properly assess the glutaminolysis dependency of the cells for the measured readouts. Also, CD138⁺ in developing PCs goes hand in hand with decreased B220 expression. A representative FACS plot showing the gating strategy for the in vitro PCs should be added as a supplementary figure. Similarly, division number (going all the way to #7) may be tricky to gate and interpret. A representative FACS plot showing the separation of B cells according to their division numbers and a subsequent gating of CD138 or IgG1 in these gates would be ideal for demonstrating the authors' ability to distinguish these populations effectively.

(3) A brief explanation should be provided for the exclusive use of IgG1 as the readout in class-switching assays, given that naïve B cells are capable of switching to multiple isotypes. Clarifying why IgG1 was preferentially selected would aid in the interpretation of the results.

(4) The immunization experiments presented in Figures 1 and 2 are well designed, and the data are comprehensively presented. However, to prevent potential misinterpretation, it should be clarified that the observed differences between NP and OVA immunizations cannot be attributed solely to the chemical nature of the antigens - hapten versus protein. A more significant distinction lies in the route of administration (intraperitoneal vs. intranasal) and the resulting anatomical compartment of the immune response (systemic vs. lung-restricted). This context should be explicitly stated to avoid overinterpretation of the comparative findings.

(5) NP immunization is known to be an inducer of an IgG1-dominant Th2-type immune response in mice. IgG2c is not a major player unless a nanoparticle delivery system is used. However, the authors arbitrarily included IgG2c in their assays in Figures 2 and 3. This may be confusing for the readers. The authors should either justify the IgG2c-mediated analyses or remove them from the main figures. (It can be added as supplemental information with proper justification).

(6) Similarly, in affinity maturation analyses, including IgM is somewhat uncommon. I do not see any point in showing high affinity (NP2/NP20) IgMs (Figure 3d), since that data probably does not mean much.

(7) Following on my comment for the PC generation in Figure 1 (see above), in Figure 4, a strategy that relies solely on CD40L stimulation is performed. This is highly artificial for the PC generation and needs to be justified, or more physiologically relevant PC generation strategies involving anti-BCR, CD40L, and various cytokines should be shown.

(8) The effects of CB839 and UK5099 on cell viability are not shown. Including viability data under these treatment conditions would be a valuable addition to the supplementary materials, as it would help readers more accurately interpret the functional outcomes observed in the study.

(9) It is not clear how the RNA seq analysis in Figure 4h was generated. The experimental strategy and the setup need to be better explained.

<https://doi.org/10.7554/eLife.107627.1.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors investigate the functional requirements for glutamine and glutaminolysis in antibody responses. The authors first demonstrate that the concentrations of glutamine in lymph nodes are substantially lower than in plasma, and that at these levels, glutamine is limiting for plasma cell differentiation *in vitro*. The authors go on to use genetic mouse models in which B cells are deficient in glutaminase 1 (Gls), the glucose transporter Slc2a1, and/or mitochondrial pyruvate carrier 2 (Mpc2) to test the importance of these pathways *in vivo*.

Interestingly, deficiency of Gl alone showed clear antibody defects when ovalbumin was used as the immunogen, but not the hapten NP. For the latter response, defects in antibody titers and affinity were observed only when both Gl and either Mpc2 or Slc2a1 were deleted. These latter findings form the basis of the synthetic auxotrophy conclusion. The authors go on to test these conclusions further using *in vitro* differentiations, Seahorse assays, pharmacological inhibitors, and targeted quantification of specific metabolites and amino acids. Finally, the authors document reduced STAT3 and STAT1 phosphorylation in response

to IL-21 and interferon (both type 1 and 2), respectively, when both glutaminolysis and mitochondrial pyruvate metabolism are prevented.

Strengths:

- (1) The main strength of the manuscript is the overall breadth of experiments performed. Orthogonal experiments are performed using genetic models, pharmacological inhibitors, in vitro assays, and in vivo experiments to support the claims. Multiple antigens are used as test immunogens--this is particularly important given the differing results.
- (2) B cell metabolism is an area of interest but understudied relative to other cell types in the immune system.
- (3) The importance of metabolic flexibility and caution when interpreting negative results is made clear from this study.

Weaknesses:

- (1) All of the in vivo studies were done in the context of boosters at 3 weeks and recall responses 1 week later. This makes specific results difficult to interpret. Primary responses, including germinal centers, are still ongoing at 3 weeks after the initial immunization. Thus, untangling what proportion of the defects are due to problems in the primary vs. memory response is difficult.
- (2) Along these lines, the defects shown in Figure 3h-i may not be due to the authors' interpretation that GLs and Mpc2 are required for efficient plasma cell differentiation from memory B cells. This interpretation would only be correct if the absence of GLs/Mpc2 leads to preferential recruitment of low-affinity memory B cells into secondary plasma cells. The more likely interpretation is that ongoing primary germinal centers are negatively impacted by GLs and Mpc2 deficiency, and this, in turn, leads to reduced affinities of serum antibodies.
- (3) The gating strategies for germinal centers and memory B cells in Supplemental Figure 2 are problematic, especially given that these data are used to claim only modest and/or statistically insignificant differences in these populations when GLs and Mpc2 are ablated. Neither strategy shows distinct flow cytometric populations, and it does not seem that the quantification focuses on antigen-specific cells.
- (4) Along these lines, the conclusions in Figure 6a-d may need to be tempered if the analysis was done on polyclonal, rather than antigen-specific cells. Alum induces a heavily type 2-biased response and is not known to induce much of an interferon signature. The authors' observations might be explained by the inclusion of other ongoing GCs unrelated to the immunization.

<https://doi.org/10.7554/eLife.107627.1.sa2>

Reviewer #3 (Public review):

Summary:

In their manuscript, the authors investigate how glutaminolysis (GLS) and mitochondrial pyruvate import (MPC2) jointly shape B cell fate and the humoral immune response. Using inducible knockout systems and metabolic inhibitors, they uncover a "synthetic auxotrophy": When GLS activity/glutaminolysis is lost together with either GLUT1-mediated glucose uptake or MPC2, B cells fail to upregulate mitochondrial respiration, IL 21/STAT3 and IFN/STAT1 signaling is impaired, and the plasma cell output and antigen-specific antibody titers drop significantly. This work thus demonstrates the promotion of plasma cell differentiation and cytokine signaling through parallel activation of two metabolic pathways. The dataset is

technically comprehensive and conceptually novel, but some aspects leave the in vivo and translational significance uncertain.

Strengths:

- (1) Conceptual novelty: the study goes beyond single-enzyme deletions to reveal conditional metabolic vulnerabilities and fate-deciding mechanisms in B cells.
- (2) Mechanistic depth: the study uncovers a novel "metabolic bottleneck" that impairs mitochondrial respiration and elevates ROS, and directly ties these changes to cytokine-receptor signaling. This is both mechanistically compelling and potentially clinically relevant.
- (3) Breadth of models and methods: inducible genetics, pharmacology, metabolomics, Seahorse assay, ELISpot/ELISA, RNA-seq, two immunization models.
- (4) Potential clinical angle: the synergy of CB839 with UK5099 and/or hydroxychloroquine hints at a druggable pathway targeting autoantibody-driven diseases.

Weaknesses:

(1) Physiological relevance of "synthetic auxotrophy"

The manuscript demonstrates that GLS loss is only crippling when glucose influx or mitochondrial pyruvate import is concurrently reduced, which the authors name "synthetic auxotrophy". I think it would help readers to clarify the terminology more and add a concise definition of "synthetic auxotrophy" versus "synthetic lethality" early in the manuscript and justify its relevance for B cells.

While the overall findings, especially the subset specificity and the clinical implications, are generally interesting, the "synthetic auxotrophy" condition feels a little engineered. Therefore, the findings strongly raise the question of the likelihood of such a "double hit" in vivo and whether there are conditions, disease states, or drug regimens that would realistically generate such a "bottleneck". Hence, the authors should document or at least discuss whether GC or inflamed niches naturally show simultaneous downregulation/lack of glutamine and/or pyruvate. The authors should also aim to provide evidence that infections (e.g., influenza), hypoxia, treatments (e.g., rapamycin), or inflammatory diseases like lupus co-limit these pathways.

It would hence also be beneficial to test the CB839 + UK5099/HQC combinations in a short, proof-of-concept treatment in vivo, e.g., shortly before and after the booster immunization or in an autoimmune model. Likewise, it may also be insightful to discuss potential effects of existing treatments (especially CB839, HCQ) on human memory B cell or PC pools.

(2) Cell survival versus differentiation phenotype

Claims that the phenotypes (e.g., reduced PC numbers) are "independent of death" and are not merely the result of artificial cell stress would benefit from Annexin-V/active-caspase 3 analyses of GC B cells and plasmablasts. Please also show viability curves for inhibitor-treated cells.

(3) Subset specificity of the metabolic phenotype

Could the metabolic differences, mitochondrial ROS, and membrane-potential changes shown for activated pan-B cells (Figure 5) also be demonstrated ex vivo for KO mouse-derived GC B cells and plasma cells? This would also be insightful to investigate following NP-immunization (e.g., NP+ GC B cells 10 days after NP-OVA immunization).

(4) Memory B cell gating strategy

I am not fully convinced that the memory-B-cell gate in Supplementary Figure 2d is appropriate. The legend implies the population is defined simply as CD19+GL7-CD38+ (or CD19+CD38++?), with no further restriction to NP-binding cells. Such a gate could also capture naïve or recently activated B cells. From the descriptions in the figure and the figure legend, it is hard to verify that the events plotted truly represent memory B cells. Please clarify the full gating hierarchy and, ideally, restrict the MBC gate to NP+CD19+GL7-CD38+ B cells (or add additional markers such as CD80 and CD273). Generally, the manuscript would benefit from a more transparent presentation of gating strategies.

(5) Deletion efficiency

mRNA data show residual GLS/MPC2 transcripts (Supplementary Figure 8). Please quantify deletion efficiency in GC B cells and plasmablasts.

<https://doi.org/10.7554/eLife.107627.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

In this manuscript, Cho et al. present a comprehensive and multidimensional analysis of glutamine metabolism in the regulation of B cell differentiation and function during immune responses. They further demonstrate how glutamine metabolism interacts with glucose uptake and utilization to modulate key intracellular processes. The manuscript is clearly written, and the experimental approaches are informative and well-executed. The authors provide a detailed mechanistic understanding through the use of both in vivo and in vitro models. The conclusions are well supported by the data, and the findings are novel and impactful. I have only a few, mostly minor, concerns related to data presentation and the rationale for certain experimental choices.

Detailed Comments:

(1) In Figure 1b, it is unclear whether total B cells or follicular B cells were used in the assay. Additionally, the in vitro class-switch recombination and plasma cell differentiation experiments were conducted without BCR stimulation, which makes the system appear overly artificial and limits physiological relevance. Although the effects of glutamine concentration on the measured parameters are evident, the results cannot be confidently interpreted as true plasma cell generation or IgG1 class switching under these conditions. The authors should moderate these claims or provide stronger justification for the chosen differentiation strategy. Incorporating a parallel assay with anti-BCR stimulation would improve the rigor and interpretability of these findings.

We will edit the manuscript to be more explicit that total splenic B cells were used in this set-up figure and the rest of the paper. In addition, we will try to perform new experiments to improve this "set-up figure" (and add old and new data for Supplemental Figure presentation). Specifically, we will increase the range of conditions tested - e.g., styles of stimulating proliferation and differentiation - to foster an increased sense of generality. We plan to compare mitogenic stimulation with anti-CD40 to anti-IgM and to anti-IgM + anti-CD40, all with BAFF, IL-4, and IL-5, bearing in mind excellent work from Aiba et al, Immunity 2006; 24: 259-268, and similar papers. We also will try to present some representative flow cytometric profiles (presumably in new Supplemental Figure panels).

To be transparent and add to a more open public discussion (using the virtues of this forum, the senior author and colleagues would caution about whether any in vitro conditions exist that warrant complete confidence. That is the reason for proceeding to immunization experiments in vivo. That is not said to cast doubt on our own in vitro data - there are some experiments (such as those of Fig. 1a-c and associated Supplemental Fig. 1) that only can be done in vitro or are better done that way (e.g., because of rapid uptake of early apoptotic B cells in vivo).

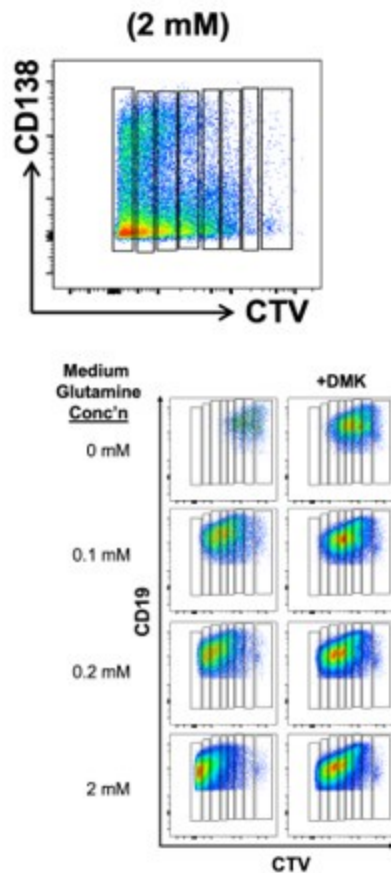
For instance: Well-respected papers use the CD40LB and NB21.2D9 systems to activate B cells and generate plasma cells. Those appear to be BCR-independent and unfortunately, we found that they cannot be used with a.a. deprivation or these inhibitors due to effects on the engineered stroma-like cells. In considering BCR engagement, Reth has published salient points about signaling and concentrations of the Ab, the upshot being that this means of activating mitogenesis and plasma cell differentiation (when the B cells are costimulated via CD40 or TLR(4 or 7/8) is probably more than a bit artificial. Moreover, although Aiba et al, Immunity 2006; 24: 259-268 is a laudable exception, one rarely finds papers using BAFF despite the strong evidence it is an essential part of the equation of B cell regulation in vivo and a cytokine that modulates BCR signaling - in the cultures.

(2) In Figure 1c, the DMK alone condition is not presented. This hinders readers' ability to properly assess the glutaminolysis dependency of the cells for the measured readouts. Also, CD138+ in developing PCs goes hand in hand with decreased B220 expression. A representative FACS plot showing the gating strategy for the in vitro PCs should be added as a supplementary figure. Similarly, division number (going all the way to #7) may be tricky to gate and interpret. A representative FACS plot showing the separation of B cells according to their division numbers and a subsequent gating of CD138 or IgG1 in these gates would be ideal for demonstrating the authors' ability to distinguish these populations effectively.

We agree that exact placement of divisions deconvolution by FlowJo is more fraught than might be thought for presentations in many or most papers. For the revision, we will try to add one or several representative FACS plot(s) with old and new data to provide the gating on CTV fluorescence, bearing these points in mind when extending the experiments from ~7 years ago (Fig. 1b, c). With the representative examples of the old data pasted in here, we will aver, however, that using divisions 0-6, and ≥ 7 was reasonable.

Ditto for DMK with normal glutamine. However, in the spirit of eLife transparency lacking in many other journals, this comparison is more fraught than the referee comment would make things seem. The concentration tolerated by cells is highly dependent on the medium and glutamine concentration, and perhaps on rates of glutaminolysis (due to its generation of ammonia). In practice, we find that DMK becomes more toxic to B cells unless glutamine is low or glutaminolysis is restricted. Thus, the concentration of DMK that is tolerated and used in Fig. 1b, c can become toxic to the B cells when using the higher levels of glutamine in typical culture media (2 mM or more) - at which point the "normal conditions + DMK" "control" involves the surviving cells in conditions with far greater cell death and less population expansion than the "low glutamine + DMK". condition. Overall, we appreciate the suggestion to show more DMK data and will work to do so for the earlier proliferation data (shown above) and the new experiments.

Author response image 1.



(3) A brief explanation should be provided for the exclusive use of IgG1 as the readout in class-switching assays, given that naïve B cells are capable of switching to multiple isotypes. Clarifying why IgG1 was preferentially selected would aid in the interpretation of the results.

We will edit the text to be more explicit and harmonize in light of the referee's suggestion that we focus the presentation of serologic data on IgG1 in the immunization experiments.

[IgG1 provides the strongest signal and hence better signal/noise both in vitro and with the alum-based immunizations that are avatars for the adjuvant used in the majority of protein-based vaccines for humans.]

(4) The immunization experiments presented in Figures 1 and 2 are well designed, and the data are comprehensively presented. However, to prevent potential misinterpretation, it should be clarified that the observed differences between NP and OVA immunizations cannot be attributed solely to the chemical nature of the antigens - hapten versus protein. A more significant distinction lies in the route of administration (intraperitoneal vs. intranasal) and the resulting anatomical compartment of the immune response (systemic vs. lung-restricted). This context should be explicitly stated to avoid overinterpretation of the comparative findings.

We agree with the referee and will edit the text accordingly. Certainly, the difference in how the anti-ova response is elicited compared to the anti-NP response in the same mice or with a

bit different an immunization regimen might be another factor - or the major factor - that could contribute towards explaining why glutaminolysis was important after ovalbumin inhalations (used because emergence of anti-ova Ab / ASCs is suppressed by the NP hapten after NP-ova immunization) but not needed for the anti-NP response unless Slc2a1 or Mpc2 also was inactivated. Thank you prompting addition of this caveat.

Nevertheless, it seems fair to note that in Figures 1 and 2, the ASCs and Ab are being analyzed for NP and ova in the same mice, albeit with the NP-specific components not being driven by the inhalations of ovalbumin. With that in mind, when one compares the IgG1 anti-NP ASC and Ab to those for IgG1 anti-ovalbumin (ASC in bone marrow; Ab), the ovalbumin-specific response was reduced whereas the anti-NP response was not.

(5) NP immunization is known to be an inducer of an IgG1-dominant Th2-type immune response in mice. IgG2c is not a major player unless a nanoparticle delivery system is used. However, the authors arbitrarily included IgG2c in their assays in Figures 2 and 3. This may be confusing for the readers. The authors should either justify the IgG2c-mediated analyses or remove them from the main figures. (It can be added as supplemental information with proper justification).

We will rearrange the Figure panels to move the IgM and IgG2c data to Supplemental Figures.

For purposes of public discourse, we note that the data of previous Figure 3(c, g) show a very strong NP-specific IgG2c response that seems to contradict the concept that IgG2c responses necessarily are weak in this setting, and the important role of IgG2c (mouse - IgG1 in humans) in controlling or clearing various pathogens as well as in autoimmunity. So from the standpoint of providing a better sense of generality to the loss-of-function effects, we continue to think that these measurements are quite important. That said, the main text has many figure panels and as the review notes, the class switching and in vitro ASC generation were done with IL-4 / IgG1-promoting conditions. If possible, we will try to assay in vitro class switching with IFN- γ rather than IL-4 but there may not be enough resources (time before lab closure; money).

[As a collegial aside, we speculate that a greater or lesser IgG2c anti-NP response may arise due to different preparations of NP-carrier obtained from the vendor (Biosearch) having different amounts of TLR (e.g., TLR4) ligand. In any case, the points of presenting the IgG2c (and IgM) data were to push against the limiting boundaries of convention (which risks perpetuating a narrow view of potential outcomes) and make the breadth of results more apparent to readers.

(6) Similarly, in affinity maturation analyses, including IgM is somewhat uncommon. I do not see any point in showing high affinity (NP2/NP20) IgMs (Figure 3d), since that data probably does not mean much.

As noted in the reply immediately preceding this one, we appreciate this suggestion from the reviewer and will move the IgM and IgG2c to Supplemental status.

Nonetheless, in collegial discourse we disagree a bit with the referee in light of our data as well as of work that (to our minds) leads one to question why inclusion of affinity maturation of IgM is so uncommon - as the referee accurately notes. Of course a defect in the capacity to class-switch is highly deleterious in patients but that is not the same as concluding that recall IgM or its affinity is of little consequence.

In some of the pioneering work back in the 1980's, Bothwell showed that NP-carrier immunization generated hybridomas producing IgM Ab with extensive SHM (~11% of the 18 lineages; ~ 1/3 of the IgM hybridomas) [PMID: 8487778], IgM B cells appear to move into GC, and there is at least a reasonable published basis for the view that there are GC-derived IgM

(unswitched) memory B cells (MBC) that would be more likely, upon recall activation, to differentiate into ASCs. [As an example, albeit with the Jenkins lab anti-rPE response, Taylor, Pape, and Jenkins generated quantitative estimates of the numbers of Ag-specific IgM⁺ vs switched MBC that were GC-derived (or not). [PMID: 22370719]. While they emphasized that ~90% of IgM⁺ MBC appeared to be GC-independent, their data also indicated that ~1/2 of all GC-derived MBC were IgM⁺ rather than switched (their Fig. 8, B vs C; also 8E, which includes alum-PE). And while we immensely respect the referee, we are perhaps less confident that IgM or high-affinity Ag-specific IgM doesn't mean that much, if only because of evidence that localized Ab compete for Ag and may thus influence selective processes [PMCID: PMC2747358; PMID: 15953185; PMID: 23420879; PMID: 27270306].

(7) Following on my comment for the PC generation in Figure 1 (see above), in Figure 4, a strategy that relies solely on CD40L stimulation is performed. This is highly artificial for the PC generation and needs to be justified, or more physiologically relevant PC generation strategies involving anti-BCR, CD40L, and various cytokines should be shown.

In line with our response to point (1), we plan and will try to self-fund testing BCR-stimulated B cells (anti-CD40 to anti-IgM and to anti-IgM + anti-CD40, all with BAFF, IL-4, and IL-5).

(8) The effects of CB839 and UK5099 on cell viability are not shown. Including viability data under these treatment conditions would be a valuable addition to the supplementary materials, as it would help readers more accurately interpret the functional outcomes observed in the study.

We will add to the supplemental figures to present data that provide cues as to relative viability / survival under the experimental conditions used. [FSC X SSC as well as 7AAD or Ghost dye panels; we also hope to generate new data that include further experiments scoring annexin V staining.]

(9) It is not clear how the RNA seq analysis in Figure 4h was generated. The experimental strategy and the setup need to be better explained.

The revised manuscript will include more information (at minimum in the Methods, Legend), and we apologize that in this and a few other instances sufficiency of detail was sacrificed on the altar of brevity.

[Adding a brief synopsis to any reader before the final version of record, given the many months it will take to generate new data, thoroughly revise the manuscript, etc:

In three temporally and biologically independent experiments, cultures were harvested 3.5 days after splenic B cells were purified and cultured as in the experiments of Fig. 4a-e. total cellular RNA prepared from the twelve samples (three replicates for each of four conditions - DMSO vehicle control, CB839, UK5099, and CB839 + UK5099) was analyzed by RNA-seq. After the RNA-seq data were initially processed using the pipeline described in the Methods. For panels g & h of Fig 4, DE Seq2 was used to quantify and compare read counts in the three CB839 + UK5099 samples relative to the three independent vehicle controls and identify all genes for which variances yielded P<0.05. In Fig 4g, all such genes for which the difference was 'statistically significant' (i.e., P<0.05) were entered into the Immgen tool and thereby mapped to the B lineage subsets shown in the figure panels (i.e., g, h). In (g), these are displayed using one format, whereas (h) uses the 'heatmap' tool in MyGeneSet.

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Strengths:

(1) The main strength of the manuscript is the overall breadth of experiments performed. Orthogonal experiments are performed using genetic models, pharmacological inhibitors, in vitro assays, and in vivo experiments to support the claims. Multiple antigens are used as test immunogens--this is particularly important given the differing results.

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Weaknesses:

(1) All of the in vivo studies were done in the context of boosters at 3 weeks and recall responses 1 week later. This makes specific results difficult to interpret. Primary responses, including germinal centers, are still ongoing at 3 weeks after the initial immunization. Thus, untangling what proportion of the defects are due to problems in the primary vs. memory response is difficult.

(2) Along these lines, the defects shown in Figure 3h-i may not be due to the authors' interpretation that Glis and Mpc2 are required for efficient plasma cell differentiation from memory B cells. This interpretation would only be correct if the absence of Glis/Mpc2 leads to preferential recruitment of low-affinity memory B cells into secondary plasma cells. The more likely interpretation is that ongoing primary germinal centers are negatively impacted by Glis and Mpc2 deficiency, and this, in turn, leads to reduced affinities of serum antibodies

We provisionally plan to edit the wording of the conclusion a bit to add a possibility we consider unlikely to avoid a conclusion that MBCs bearing switched BCRs are affected once reactivated. We also will perform a new experiment to investigate, but unfortunately time before lab closure has been and remains our enemy both for performance and multiple replication of the work presented in Figure 3, panels h & i, and the related Supplemental Data (Supplemental Fig. 3a-j). Unfortunately, it will not be possible to do a memory experiment with recall immunization out at 8 weeks. Despite the grant funding running out and

institutional belt-tightening, however, we'll try to perform a new head-to-head comparison of 4 wk post-immunization with and without the boost at three weeks.

The intriguing concern (points 1 & 2) provides a springboard for consideration of generalizations and simplifications. Germinal center durability is not at all monolithic, and instead is quite variable**. The premise (cognitive bias, perhaps?) in the interpretation is that in our previous work we find few if any GC B cells - NP-APC-binding or otherwise - above the background (non-immunized controls) three weeks after immunization with NP-ovalbumin in alum. Recognizing that it is not NP-carrier in alum as immunizations, we note for the readers and referee that Fig. 1 of the Taylor, Pape, & Jenkins paper considered above [PMID: 22370719] reported 10-fold more Ag-specific MBCs than GC B cells at day 29 post-immunization (the point at which the boost / recall challenge was performed in our Figure 3h, i).

Viewed from that perspective, the surmise of the comment is that a major contribution to the differences in both all-affinity and high-affinity anti-NP IgG1 shown in Fig. 3i derives from the immunization at 4 wk stimulating GC B cells we cannot find as opposed to memory B cells. However, it is true that in the literature (especially with the experimentally different approach of transferring BCR-transgenic / knock-in versions of an NP-biased BCR) there may be meaningful pools of IgG1 and IgG2c GC B cells. Alternatively, our current reagents for immunizations may have become better at maintaining GC than those in the past - which we will try to test.

The issue and question also relate to rates of output of plasma cells or rises in the serum concentrations of class-switched Ab. To this point, our prior experiences agree with the long-published data of the Kurosaki lab in Figure 3c of the Aiba et al paper noted above (Immunity, 2006) (and other such time courses). Readers can note that the IgG1 anti-NP response (alum adjuvant, as in our work) hits its plateau at 2 wk, and did not increase further from 2 to 3 wk. In other words, GC are on the decline and Ab production has reached its plateau by the time of the 2nd immunization in Fig. 3h).

Assuming we understand the comment and line of reasoning correctly, we also lean towards disagreeing with the statement "This interpretation would only be correct if the absence of GLs/Mpc2 leads to preferential recruitment of low-affinity memory B cells into secondary plasma cells." Our evidence shows that both low-affinity as well as high-affinity anti-NP Ab (IgG1) went down as a result of combined gene-inactivation after the peak primary response (Fig. 3i). Recent papers show that affinity maturation is attributable to greater proliferation of plasmablasts with high-affinity BCR. Accordingly, the findings with loss of GLS and MPC function are quite consistent with the interpretation that much of the response after the second immunization draws on MBC differentiation into plasmablasts and then plasma cells, where the proliferative advantage of high-affinity cells is blunted by the impaired metabolism. The provisional plan, however, is to note the alternative, if less likely, interpretation proposed by the review.

** In some contexts, of course, especially certain viral infections or vaccination with lipid nanoparticles carrying modified mRNA, germinal centers are far more persistent; also, in humans even the seasonal flu vaccine **

(3) The gating strategies for germinal centers and memory B cells in Supplemental Figure 2 are problematic, especially given that these data are used to claim only modest and/or statistically insignificant differences in these populations when GLs and Mpc2 are ablated. Neither strategy shows distinct flow cytometric populations, and it does not seem that the quantification focuses on antigen-specific cells.

We will enhance these aspects of the presentation, using old and hopefully new data, but note for readers that many many other papers in the best journals show plots in which the

separation of, say, GC-Tfh from overall Tfh is based on cut-off within what essentially is a continuous spectrum of emission as adjusted or compensated by the cytometer (spectral or conventional).

Perhaps incorrectly, we omitted presenting data that included the results with NP-APC-staining - in part because within the GC B cell gate the frequencies of NP-binding events (GCB cells) were similar in double-knockout samples and controls. In practice, that would mean that the metabolic requirement applied about equally to NP+ and the total population. We will try to rectify this point in the revision.

(4) Along these lines, the conclusions in Figure 6a-d may need to be tempered if the analysis was done on polyclonal, rather than antigen-specific cells. Alum induces a heavily type 2-biased response and is not known to induce much of an interferon signature. The authors' observations might be explained by the inclusion of other ongoing GCs unrelated to the immunization.

We will make sure the text is clear that the in vitro experiments do not represent GC B cells and that the RNA-seq data were not an Ag (SRBC)-specific subset.

We also will try to work in a schematic along with expanding the Legends to make it more readily clear that the RNA-seq data (and hence the GSEA) involved immunizations with SRBC (not the alum / NP system which - it may be noted - in these experiments actually generated a robust IgG2c (type 1-driven) response along with the type 2-enhanced IgG1 response.

Reviewer #3 (Public review):

Summary:

In their manuscript, the authors investigate how glutaminolysis (GLS) and mitochondrial pyruvate import (MPC2) jointly shape B cell fate and the humoral immune response. Using inducible knockout systems and metabolic inhibitors, they uncover a "synthetic auxotrophy": When GLS activity/glutaminolysis is lost together with either GLUT1-mediated glucose uptake or MPC2, B cells fail to upregulate mitochondrial respiration, IL 21/STAT3 and IFN/STAT1 signaling is impaired, and the plasma cell output and antigen-specific antibody titers drop significantly. This work thus demonstrates the promotion of plasma cell differentiation and cytokine signaling through parallel activation of two metabolic pathways. The dataset is technically comprehensive and conceptually novel, but some aspects leave the in vivo and translational significance uncertain.

Strengths:

(1) Conceptual novelty: the study goes beyond single-enzyme deletions to reveal conditional metabolic vulnerabilities and fate-deciding mechanisms in B cells.

(2) Mechanistic depth: the study uncovers a novel "metabolic bottleneck" that impairs mitochondrial respiration and elevates ROS, and directly ties these changes to cytokine-receptor signaling. This is both mechanistically compelling and potentially clinically relevant.

(3) Breadth of models and methods: inducible genetics, pharmacology, metabolomics, Seahorse assay, ELISpot/ELISA, RNA-seq, two immunization models.

(4) Potential clinical angle: the synergy of CB839 with UK5099 and/or hydroxychloroquine hints at a druggable pathway targeting autoantibody-driven diseases.

We agree and thank the referee for the positive comments and this succinct summary of what we view as contributions of the paper.

Weaknesses:

(1) Physiological relevance of "synthetic auxotrophy"

The manuscript demonstrates that GLS loss is only crippling when glucose influx or mitochondrial pyruvate import is concurrently reduced, which the authors name "synthetic auxotrophy". I think it would help readers to clarify the terminology more and add a concise definition of "synthetic auxotrophy" versus "synthetic lethality" early in the manuscript and justify its relevance for B cells.

We will edit the Abstract, Introduction, and Discussion to try to do better on this score. Conscious of how expansive the prose and data are even in the original submission, we appear to have taken some shortcuts that we will try to rectify. Thank you for highlighting this need to improve on a key concept!

That said, we punctiliously & perhaps pedantically encourage readers to be completely accurate, in that under one condition of immunization GLS loss substantially reduced the anti-ovalbumin response (Fig. 1, Fig. 2a-c). And for this provisional response, we will expand a bit on the notion that synthetic auxotrophy represents effects on differentiation that appear to go beyond and not simply to be selective death, even though decreased population expansion is observed and one cannot exclude some contribution of enhanced death in vivo. Finally, we will note that this comment of the review raises interesting semantic questions about what represents "physiological relevance" but leave it at that.

While the overall findings, especially the subset specificity and the clinical implications, are generally interesting, the "synthetic auxotrophy" condition feels a little engineered.

One can readily say that CAR-T cells are 'a little engineered' so it is a matter of balancing this perspective of the referee against the strengths they highlight in points 1, 2, and 4. In any case, we will probably try to expand and be more explicit in the Discussion of the revised manuscript.

In brief, even were the money not all gone, we would not believe that expanding the heft of this already rather large manuscript and set of data would be appropriate. As matters stand, a basic new insight about metabolic flexibility and its limits leads to evidence of a way to reduce generation of Ab and a novel impairment of STAT transcription factor induction by several cytokine receptors. The vulnerability that could be tested in later work on B cell-dependent autoimmunity includes the capacity to test a compound that already has been to or through FDA phase II in patients together with an FDA-approved standard-of-care agent.

Put a different way, the point is that a basic curiosity to understand why decreasing glucose influx did not have an even more profound effect than what was observed, combined with curiosity as to why glutaminolysis was dispensable in relatively standard vaccine-like models of immunize / boost, provided a springboard to identification of new vulnerabilities. As above, we appreciate being made aware that this point merits being made more explicit in the Discussion of the edited version.

Therefore, the findings strongly raise the question of the likelihood of such a "double hit" in vivo and whether there are conditions, disease states, or drug regimens that would realistically generate such a "bottleneck".

Hence, the authors should document or at least discuss whether GC or inflamed niches naturally show simultaneous downregulation/lack of glutamine and/or pyruvate. The authors should also aim to provide evidence that infections (e.g., influenza), hypoxia,

treatments (e.g., rapamycin), or inflammatory diseases like lupus co-limit these pathways.

Again, we appreciate some 'licensing' to be more expansive and explicit, and will try to balance editing in such points against undue tedium or tendentiously speculative length in the Discussion. In particular, we will note that a clear, simple implication of the work is to highlight an imperative to test CB839 in lupus patients already on hydroxychloroquine as standard-of-care, and to suggest development of UK5099 (already tested many times in mouse models of cancer) to complement glutaminase inhibition.

As backdrop, we note that the failure to advance imaging mass spectrometry to the capacity to quantify relative or absolute (via nano-DESI) concentrations of nutrients in localized interstitia is a critical gap in the entire field. Techniques that sample the interstitial fluid of tumor masses or in our case LN as a work-around have yielded evidence that there can be meaningful limitations of glucose and glutamine, but it needs to be acknowledged that such findings may be very model-specific and, as can be the case with cutting-edge science, are not without controversy. That said, yes, we had found that hypoxia reduced glutamine uptake but given the norms of focused, tidy packages only reported on leucine in an earlier paper [PMID27501247; PMID5161594].

It would hence also be beneficial to test the CB839 + UK5099/HCQ combinations in a short, proof-of-concept treatment in vivo, e.g., shortly before and after the booster immunization or in an autoimmune model. Likewise, it may also be insightful to discuss potential effects of existing treatments (especially CB839, HCQ) on human memory B cell or PC pools.

We certainly agree that the suggestions offered in this comment are important next steps and the right approach to test if the findings reported here translate toward the treatment of autoimmune diseases that involve B cells, interferons, and pathophysiology mediated by auto-Ab. As practical points, performance and replication of such studies would take more time than the year allotted for return of a revised manuscript to eLife and in any case neither funds nor a lab remain to do these important studies.

Concrete evidence for our concurrence was embodied in a grant application to NIH that was essential for keeping a lab and doing any such studies. [We note, as a suggestion to others, that an essential component of such studies would be to test the effects of these compounds on B cells from patients and mice with autoimmunity]. Perhaps unfortunately for SLE patients, the review panelists did not agree about the importance of such studies. However, it can be hoped that the patent-holder of CB839 (and perhaps other companies developing glutaminase inhibitors) will see this peer-reviewed pre-print and the public dialogue, and recognize how positive results might open a valuable contribution to mitigation of diseases such as SLE.

(2) Cell survival versus differentiation phenotype

Claims that the phenotypes (e.g., reduced PC numbers) are "independent of death" and are not merely the result of artificial cell stress would benefit from Annexin-V/active-caspase 3 analyses of GC B cells and plasmablasts. Please also show viability curves for inhibitor-treated cell

This comment leads us to see that the wording on this point may have been overly terse in the interests of brevity, and thereby open to some misunderstanding. Accordingly, we will expand out the text of the Abstract and elsewhere in the manuscript, to be more clear. In addition, we will add in some data on the point, hopefully including some results of new experiments.

To clarify in this public context, it is not that an increase in death (along with the reported decrease in cell cycling) can be or is excluded - and in fact it likely exists in vitro. The point is that beyond any such increase, and taking into account division number (since there is evidence that PC differentiation and output numbers involve a 'division-counting' mechanism), the frequencies of CD138⁺ cells and of ASCs among the viable cells are lower, as is the level of Prdm1-encoded mRNA even before the big increase in CD138⁺ cells in the population.

(3) Subset specificity of the metabolic phenotype

Could the metabolic differences, mitochondrial ROS, and membrane-potential changes shown for activated pan-B cells (Figure 5) also be demonstrated ex vivo for KO mouse-derived GC B cells and plasma cells? This would also be insightful to investigate following NP-immunization (e.g., NP⁺ GC B cells 10 days after NP-OVA immunization).

We agree that such data could be nice and add to the comprehensiveness of the work. We will try to scrounge the resources (time; money; human) to test this roughly as indicated. That said, we would note that the frequencies and hence numbers of NP⁺ GC B cells are so low that even in the flow cytometer we suspect there will not be enough "events" to rely on the results with DCFDA in the tiny sub-sub-subset. It also bears noting that reliable flow cytometric identification of the small NP-specific plasmablast/plasma cell subset amidst the overall population, little of which arose from immunization or after deletion of the floxed segments in B cells, would potentially be misleading.

(4) Memory B cell gating strategy

I am not fully convinced that the memory-B-cell gate in Supplementary Figure 2d is appropriate. The legend implies the population is defined simply as CD19⁺GL7⁺CD38⁺ (or CD19⁺CD38^{++?}), with no further restriction to NP-binding cells. Such a gate could also capture naïve or recently activated B cells. From the descriptions in the figure and the figure legend, it is hard to verify that the events plotted truly represent memory B cells. Please clarify the full gating hierarchy and, ideally, restrict the MBC gate to NP⁺CD19⁺GL7⁺CD38⁺ B cells (or add additional markers such as CD80 and CD273). Generally, the manuscript would benefit from a more transparent presentation of gating strategies.

We will further expand the supplemental data displays to include more of the gating and analytic scheme, and hope to be able to have performed new experiments and analyses (including additional markers) that could mitigate the concern noted here. In addition, we will include flow data from the non-immunized control mice that had been analyzed concurrently in the experiments illustrated in this Figure.

Although it should be noted that the labeling indicated that the gating included the important criterion that cells be IgD⁻ (Supplemental Fig. 2b), which excludes the vast majority of naïve B cells, in principle marginal zone (MZ) B cells might fall within this gate. However, the MZ B population is unlikely to explain the differences shown in Supplemental Fig. 2b-d.

(5) Deletion efficiency - [The] mRNA data show residual GLS/MPC2 transcripts (Supplementary Figure 8). Please quantify deletion efficiency in GC B cells and plasmablasts.

Even were there resources to do this, the degree of reduction in target mRNA (Gls; Mpc2) renders this question superfluous.

Are there likely to be some cells with only one, or even neither, allele converted from fl to D? Yes, but they would be a minor subset in light of the magnitude of mRNA reduction, in contrast to our published observations with Slc2a1. As to plasmablasts and plasma cells, the pre-existing populations make such an analysis misleading, while the scarcity of such cells recoverable with antigen capture techniques is so low as to make both RNA and genomic DNA analyses questionable.

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